De-Anonymization and Punitive Tax Enforcement in Multi-Chain Obfuscation Networks

A Technical White Paper on Advanced Blockchain Forensics and Liability Assessment

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Abstract

The proliferation of multi-chain decentralized finance (DeFi) architectures can complicate the reconstruction of capital flows across pseudonymous addresses. This white paper presents a nine-phase framework for heuristic clustering, cross-chain state representation, endpoint attribution, conditional liability assessment and constrained recovery. A unified event-level Directed Acyclic Graph (DAG) and associated tensors organize the available evidence. Statistical scores and custodial records can support attribution hypotheses, but do not automatically collapse every address into one legal identity. The subsequent sections distinguish verified mathematical identities from stipulated zero-basis assumptions, penalty and interest models, and implementation-dependent enforcement outcomes. The full derivations, examples and definitions are retained with the conditions and counterexamples identified by validation.

To facilitate reproducibility, Python validation scripts accompany the mathematical results identified in De-Anonymization and Punitive Tax Enforcement in Multi-Chain Obfuscation Networks.. The scripts implement the described algorithms and perform exact finite computations and numerical consistency checks supporting the analysis. These computational checks complement, but do not replace, the mathematical proofs and their stated assumptions. The code is available at:

https://github.com/teknoquants/De-Anonymization-and-Punitive-Tax-Enforcement-in-Multi-Chain-Obfuscation-Networks

Introduction

Decentralized finance (DeFi) architectures and multi-chain networks permit capital to move through thousands of pseudonymous addresses. Non-custodial peeling chains and cross-chain bridge smart contracts can make accounting and attribution difficult, but graph complexity alone does not establish intent or make provenance irretrievable. This white paper develops a framework for reconstructing supported transaction relationships while preserving the distinction between ledger events, statistical inferences and legal conclusions. A candidate address cluster becomes a legally attributable entity only when additional ownership and identity evidence supports that interpretation.

The text examines the transition from algorithmic attribution to conditional financial liability through heuristic clustering, cross-chain state unification and custodial records. It retains the original discussion of basis substantiation, pricing oracles, smart-contract freezing and threshold custody. The mathematics can quantify specified scenarios; it cannot itself establish accepted basis, fraud, procedural authority or certainty of total forfeiture. Corporate transfer-pricing and routing examples provide useful comparisons, but graph homomorphism, graph isomorphism and legal equivalence require different proofs. These distinctions apply throughout the nine phases, including where the original section titles identify an enforcement objective rather than an established outcome.

Revision scope. This full-length edition restores the original section sequence, proof discussions, numbered definitions, simulation table and bibliography. Statements contradicted by the validation are corrected at their original locations. Additional qualifications explain the limits of finite test fixtures and of the model assumptions. “Actor,” “target” and the source term “evader” refer to modeled roles; their use is not a finding of wrongdoing. Legal citations remain contextual references requiring independent verification.

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Phase 1: Heuristic Clustering and Cross-Chain Forensics

1.1 Proposition 1.1 Canonical Acyclic Representation under Causal and Conservation Conditions

Proposition 1.1. Let C = {C_1,...,C_M} denote asynchronously validating ledgers, each with a specified UTXO or account-state representation. Their supported state transitions within a bounded observation interval can be organized as a time-expanded graph G = (V,E,W). If every local and inter-chain edge respects one strict causal rank, G is a DAG. Per-asset conservation additionally requires opening and closing balances, carry-forward states, fees, minting and burning to be represented consistently. This establishes an acyclic analytical representation under explicit conditions, rather than an unconditional isomorphism between every ledger and the normalized graph.

Proof

Let each chain C_m be formalized as a tuple (S_m, Δ_m, Π_m), where S_m is its state space, Δ_m is the state transition function and Π_m specifies its consensus and finality policy. Use the sequence of blocks treated as finalized under that policy. The representation must retain transaction and execution order, since several state changes can occur at the same block timestamp. An account-based chain need not use Ethereum's particular trie implementation for the causal-graph argument to apply.

1. **Vertex Space Partitioning:**
    - For ledgers implementing the UTXO paradigm, define the local vertex subset as $V_{UTXO} = \{o_i \mid o_i \text{ is an unspent or consumed output generated in } \Delta_m\}$.
    - **For ledgers implementing the Account-based paradigm, let A_m be the set of persistent cryptographic addresses. Because account states are mutable across block height $h \in \mathbb{N}$ and transaction index $k \in \mathbb{N}$, define the localized vertex space as the set of discrete account-state instances:**

$$V_{ACC} = \{a_{u,h,k} \mid a_u \in A_m,\ (h,k) \text{ uniquely identifies a balance-altering mutation}\}$$

The global vertex set of the multi-chain graph is the disjoint union:

$$V = V_{UTXO} \cup V_{ACC}$$

2. **Edge Generation and Valuations:**

A directed edge e = (u,v) records a supported value transfer or state dependency from u to v. Intra-chain transfers are established from the applicable execution records. A cross-chain edge requires evidence of the bridge's actual source-to-destination dependency; it is not implied merely by similar amounts or timestamps. For transfer edges, retain a valuation tuple containing quantity, recorded time, asset identity and fee, together with unique event identifiers and the evidence supporting the edge:

$$W(e) = [\nu_e, \tau_e, \alpha_e, \phi_e]$$

Here ν_e is a nonnegative quantity in stated asset units, τ_e is the recorded timestamp rather than a guaranteed exact physical clock reading, α_e is the canonical asset identifier, and ϕ_e is the explicitly attributed protocol fee. If a fee is paid in a different asset, retain its asset label separately rather than subtracting incompatible quantities.

3. **Strict Acyclicity:**

Suppose, for contradiction, that G contains a directed cycle $\gamma = \langle v_0, v_1, \dots, v_{n-1}, v_0 \rangle$ of length $n \geq 1$.

- Case A (Intra-chain edge). For a dependency (v_i,v_(i+1)) on one ledger, use block, transaction and execution ordering to assign a strict causal rank r(v_i) < r(v_(i+1)). This ordering does not require distinct recorded timestamps. The original argument's assumed positive lower bound on timestamp differences was unnecessary and does not hold for all events within a block.

- Case B (Inter-chain edge). A bridge dependency must be verified under its actual trust and finality mechanism. Some systems use proofs of source finality; others have different assumptions. Add the edge only when the destination event depends on the identified source event and is consistent with the combined causal ordering. A globally consistent strict rank is a premise of the DAG construction, not something established by subtracting asynchronously recorded clock values.

Summing strict rank increments around the assumed cycle gives:

$$\sum_{i=0}^{n-1}[r(v_{i+1}) - r(v_i)] \;=\; 0 \;>\; 0$$

The left side telescopes to zero because v_n = v_0, whereas each edge would contribute a positive increment. The contradiction proves acyclicity under the strict-rank premise. Aggregating all time-expanded states of an address into one vertex can create cycles, so this conclusion does not extend automatically to an address-level graph.

4. **Weak Conservation of Value:**

Let mint and burn terms explicitly represent asset creation and destruction. At a state boundary for an asset α, conservation includes any balance retained after the observed outgoing transfers. Using a fee convention that does not count the same fee twice, the appropriate bookkeeping identity is:

$$b_{open} + in + mint \;=\; out + b_{close} + fee + burn$$

All terms in this identity use the same asset units. The original fee-only inflow-minus-outflow residual fails when an account retains a balance unless that balance is represented by a carry-forward edge. AMM slippage is a valuation effect, not automatically destruction of token units. Multi-asset swaps therefore require asset-specific accounting and a separate price model. Under these conditions, the equation supplies the intended weak conservation statement. ■

Mathematical Definition 1.1: Unified Multi-Chain Spatiotemporal Adjacency Tensor

To replace topological network schematics with a rigorous computational structure, the multi-chain graph $G = (V, E, W)$ is formalized as an order-4 tensor field over a partitioned temporal domain.

Definition 1.1.1 (Temporal Epoch Partition). The observation interval $[t_0, t_{end}]$ is partitioned into K uniform, non-overlapping temporal quanta:

$$\mathcal{T} = \{\Delta t_k = [t_0 + k \cdot \tau,\ t_0 + (k+1) \cdot \tau) \mid k \in \{0,1,\dots,K-1\}\}$$

where $\tau = \frac{t_{end} - t_0}{K}$ represents the temporal discretization resolution.

Definition 1.1.2 (Adjacency Tensor). Let |V| = N and let A be the number of asset classes. The following order-four tensor aggregates transfer amounts by source, destination, asset and time bin. It is a useful summary of the graph, but not a complete representation of its underlying event records:

$$\mathbf{A} \in \mathbb{R}^{N \times N \times A \times K}$$

whose individual components are evaluated via the path-integral indicator mapping:

$$\mathbf{A}_{i,j,\alpha,k} = \sum_{e \in E_{i \to j}} v_e \cdot \mathbb{I}(\alpha_e = \alpha) \cdot \mathbb{I}(\tau_e \in \Delta t_k)$$

where $\mathbb{I}(\cdot)$ is the discrete indicator function:

$$\mathbb{I}(P) = \begin{cases} 1 & \text{if predicate } P \text{ is true} \\ 0 & \text{otherwise} \end{cases}$$

Definition 1.1.3 (Adjacency Flow Operator). For source-row adjacency and a column source selector X, the destination flow totals for a single asset α and epoch k are computed by transposition:

$$L_{\alpha,k}(X) = A_{:,:,\alpha,k}{}^{T}X$$

This operator aggregates selected outgoing flows by destination. It is not a balance transition matrix: a balance update also subtracts outflows and accounts for retained balances, minting, burning and fees. Unlike token quantities cannot be summed across α without conversion to common value units. Retain exact event times, multiplicity and fees separately, because transfers of 7 and 3 can produce the same amount-tensor entry as one transfer of 10. The half-open partition also needs an explicit convention for an event exactly at the observation interval's upper boundary.

1.2 Lemma 1.2 Asymmetric Entropy as a Peeling Chain Change Heuristic

Lemma 1.2. Let Tx = (I,O,ϕ) be a transaction in a peeling-chain model with at least two outputs. An intended payment output o_p and a residual change output o_c may exhibit different fractional-digit patterns when payments are round and fees vary. The entropy score below is a candidate classification heuristic. Its mathematical identity is valid, but the probability that it identifies each new change output does not converge to one merely because chain depth increases. Classification accuracy requires a specified transaction model, calibrated features, labels and a tie-breaking policy.

Proof

1. **Fractional Value Representation:**

Express the atomic value $\nu(o_j)$ of any output $o_j \in O$ in terms of its fixed-point fractional decomposition:

$$\nu(o_j) = \sum_{r=-d}^{p} a_r(o_j) \cdot 10^r, \quad a_r(o_j) \in \{0,1,\dots,9\}$$

where d represents the fractional token precision (e.g., $d = 8$ for BTC, $d = 18$ for ETH).

2. **Mantissa Information Divergence:**

Define the discrete probability distribution $\mathbf{p}(o_j) = [p_0, p_1, \dots, p_9]$ of the least significant fractional digits $\{a_{-1}, a_{-2}, \dots, a_{-d}\}$:

$$p_s(o_j) = \frac{1}{d} \sum_{r=-d}^{-1} \mathbb{I}(a_r(o_j) = s)$$

Let $\mathbf{u} = \left[\frac{1}{10}, \frac{1}{10}, \dots, \frac{1}{10}\right]$ represent the uniform maximum-entropy distribution. The Kullback-Leibler (KL) divergence of output o_j relative to the uniform distribution is:

$$D_{KL}(\mathbf{p}(o_j) \parallel \mathbf{u}) = \sum_{s=0}^{9} p_s(o_j) \log\left(\frac{p_s(o_j)}{1/10}\right) = \log(10) - H(\mathbf{p}(o_j))$$

where $H(\mathbf{p}(o_j)) = -\sum_{s=0}^{9} p_s(o_j) \log(p_s(o_j))$ is the Shannon entropy.

3. **Stochastic Fee Impact on Residuals:**

Under a round-payment hypothesis, a peel amount can be configured to meet a fiat or token target, producing trailing zero digits. The following is a possible feature condition, not a property of every payment output:

$$\exists k^* \in \{-d, \dots, -1\} \quad \text{such that} \quad \forall r \le k^*,\ a_r(o_p) = 0$$

Trailing zeros can reduce digit entropy, but the distribution becomes a delta distribution only when all counted digits coincide. Non-round payments can have high digit entropy, and change outputs can themselves be round. Consequently, the score separation posited by the heuristic must be measured rather than assumed.

Conversely, the change output value is strictly constrained by the total consumed input value $\sum_{i\in I} \nu(i)$ and the consensus execution fee ϕ:

$$\nu(o_c) = \sum_{i\in I} \nu(i) - \nu(o_p) - \phi$$

For an account-based gas model, the fee can be represented as GasUsed × EffectiveGasPrice. Other ledgers require their own fee model. Approximating fees by a continuous random variable can be useful analytically, but continuity or randomness alone does not imply a uniform fractional residue of the resulting change amount.

$$\nu_{\text{change}} \bmod 10^{-k^*} : \text{ distribution requires a specified fee model}$$

An individual output has only d observed fractional digits. Even if change residues were generated from a uniform population, their finite empirical distribution would not generally equal the ten-symbol uniform distribution. With d = 8, the entropy is bounded by log(8), which gives the counterexample to the original zero-divergence limit:

$$D_{KL}(p_{\text{change}} \parallel u) \geq \log(10/8) > 0 \quad \text{when } d = 8$$

4. **Iterative Convergence:**

For an automated peeling chain of depth D, let $Y_k \in \{0, 1\}$ be the binary indicator that the change output is correctly classified at step $k \in \{1, 2, \dots, D\}$ by the decision rule:

$$o^* = \arg\min_{o_j \in O_k} D_{KL}(\mathbf{p}(o_j) \parallel \mathbf{u})$$

For distinct outputs, suppose a calibrated per-step classifier has a fixed error probability ε. Independent fee distributions do not by themselves establish independent classification errors; even if those errors were independent, observing more steps does not improve the marginal accuracy of a new output. The product below instead describes the probability of getting every one of D independent classifications correct:

$$\Pr(Y_1 = \cdots = Y_D = 1) = (1 - \varepsilon)^D$$

Thus the individual success probability remains 1−ε under that model, while the probability of an error-free sequence decreases when ε > 0. Repeated independent evidence about one common hypothesis is a different setting in which a justified likelihood model can concentrate a posterior. It cannot be substituted for inference about distinct change outputs:

$$\Pr(Y_D = 1) = 1 - \varepsilon, \quad \text{not a depth} - \text{driven limit of 1}$$

■

Mathematical Definition 1.2: Peeling-Chain Structural Transformation Operator

Rather than relying on visual graph diagrams of peeling chains, the decomposition of unspent outputs is formalized via linear transformation operators and state space projections.

Definition 1.2.1 (Input-Output Allocation Matrix). For a peeling transaction T_x with an input vector $\mathbf{v}_{in} \in \mathbb{R}^{|I|}$ and output vector $\mathbf{v}_{out} \in \mathbb{R}^{|O|}$, the allocation is defined by the transformation:

$$\mathbf{v}_{out} = \mathbf{M}_{T_x}\mathbf{v}_{in} - \boldsymbol{\phi}_{T_x}$$

where $\mathbf{M}_{T_x} \in \mathbb{R}^{|O|\times|I|}$ satisfies the stochastic mass conservation condition:

$$\sum_{j=1}^{|O|}\left(\mathbf{M}_{T_x}\right)_{j,i} = 1, \quad \forall i \in \{1, \dots, |I|\}$$

and $\boldsymbol{\phi}_{T_x} = \left[\frac{\phi}{|O|}, \dots, \frac{\phi}{|O|}\right]^T$ represents the fee dissipation vector.

Allocation conditions. Require M_Tx to be nonnegative and M_Tx v_in to exceed the allocated fee vector componentwise. Column sums of one then preserve total quantity after the fee. Equal allocation of the fee to outputs is a convention; it does not establish an intrinsic attribution of that fee to each output.

Definition 1.2.2 (Change Projection Functional). Define the structural projection functional $\boldsymbol{\Pi}_{change}$: $\mathbb{R}^{|O|} \to \mathbb{R}$ which maps the output space directly to the scalar change element:

$$\Pi_{change}(\mathbf{v}_{out}) = \mathbf{e}_c^T \mathbf{v}_{out} = v(o_c)$$

where the basis vector $\mathbf{e}_c \in \mathbb{R}^{|O|}$ is evaluated algorithmically via the decision vector:

$$\mathbf{e}_c = \arg\max_{\mathbf{e}_j}\left(\mathbf{w}^T \mathbf{F}(o_j)\right)$$

with weight vector $\mathbf{w} \in \mathbb{R}^4$ and feature transformation $\mathbf{F}$: $\boldsymbol{V} \to \mathbb{R}^4$:

$$\mathbf{F}(o_j) = \begin{bmatrix} \mathbb{I}\left(\deg_{in}^{-}(o_j) = 0\right) \\ H\left(\mathbf{p}(o_j)\right) \\ \mathbb{I}\left(\text{scriptPubKey}(o_j) = \text{scriptPubKey}(I)\right) \\ \mathbb{I}\left(\text{out_degree}(o_j) = 1\right) \end{bmatrix}$$

Projection conditions. The selected basis vector extracts a component but does not prove that it is change. Define the feature weights, encoding, measurement time and handling of ties. Future out-degree is a retrospective feature and would introduce look-ahead if claimed to be known at transaction time. Pooling digits across many outputs would likewise be a different estimator from the fixed-precision per-output distribution above.

1.3 Theorem 1.3 Cross Chain Correlation and Conditional Statistical Bounds

Theorem 1.3. *Let C_A and C_B be distinct, asynchronously executing distributed ledgers with independent state trees. Let $d \in \mathcal{D}$ denote a lock/burn event of asset α_A at block time t_d on C_A, and let $w \in \mathcal{W}$ denote a corresponding mint/release event of asset α_B at block time t_w on C_B. Assume the bridge relayer mechanism exhibits latency bounded by the probability density function $f_\Delta(t)$ with finite mean μ_t and variance σ_t^2, and that intermediate trades execute over Automated Market Maker (AMM) pools governed by invariant function $\Psi(x, y) = k$.*

The cross-chain trajectory correlation function defined by:

$$S(d, w) = \exp\left(-\frac{1}{2}\left[\left(\frac{(t_w - t_d) - \mu_t}{\sigma_t}\right)^2 + \left(\frac{v_w - \mathcal{E}(v_d, t_d)}{\sigma_v}\right)^2\right]\right)$$

is a bounded similarity score, not a normalized probability density or a distance. An exponential false-pair bound can be established for the specific acceptance region below only after adding a suitable null model. If unrelated-pair features are independent standard normal variables and $0 < \varepsilon < 1$, the corrected bound is:

$$P_{FP} \leq \exp[(K/2)(1 - \varepsilon^2 + \log(\varepsilon^2))]$$

This bound applies to the single-pair event $\|Z\|^2/K \leq \varepsilon^2$ under the stated Gaussian null. It is not valid for arbitrary $\varepsilon > 0$ or for arbitrary standardized blockchain features. A search over many candidate pairs requires an additional multiple-comparison analysis.

Proof

1. **Bridge Execution Latency Bounds:**

The elapsed time between a supported source and destination event may be modeled as a sum of source-confirmation, relayer and destination-consensus intervals. Independence among these intervals is an additional modeling assumption:

$$\Delta T = t_w - t_d = \Delta t_{confirm}(C_A) + \Delta t_{relayer} + \Delta t_{consensus}(C_B)$$

If the component intervals are bounded, their sum is bounded, but neither boundedness nor increasing the number of observed samples makes a fixed three-term sum normal. Define the standardized variable using its actual nonzero standard deviation:

$$Z_t = \frac{\Delta T - \mu_t}{\sigma_t}$$

A normal approximation for this variable needs empirical support or an applicable limit theorem with its conditions. Increasing sample size improves estimation of the existing population distribution; it does not change that population into N(0,1). A sum of three independent uniforms supplies a concrete counterexample to the original claim.

2. **Volumetric Expectation Under AMM Invariants:**

Let the bridge execute an automated liquidity rebalancing step via a constant-product AMM pool on chain C_B. The invariant condition requires:

$$\left(R_A + \nu_d \cdot \left(1 - \phi_{protocol}\right)\right)(R_B - \nu_w) = R_A R_B = k$$

Here R_A and R_B are positive supplied reserves and ϕ_protocol is a fee fraction in [0,1]. Solving the effective-input invariant gives the conditional output function denoted E in the original notation:

$$\mathcal{E}(\nu_d, t_d) = \frac{\nu_d \cdot \left(1 - \phi_{protocol}\right) \cdot R_B(t_d)}{R_A(t_d) + \nu_d \cdot \left(1 - \phi_{protocol}\right)}$$

If reserves or prices are random, this conditional output is not automatically their statistical expectation. Oracle noise, slippage and execution costs require a defined joint model. For example, zero-mean oracle noise plus Uniform(0,b) slippage has mean b/2. The uncentered volume residual below must be adjusted by its actual mean and scaled by its actual standard deviation before calling it standardized:

$$Z_\nu = \frac{\nu_w - \mathcal{E}(\nu_d, t_d)}{\sigma_\nu}$$

The properties E[Z_ν] = 0 and Var(Z_ν) = 1 therefore hold only after the required centering and scaling. They are not consequences of subtracting the AMM output alone. Under that restriction the volume and latency components can form a calibrated feature vector.

3. **Composite Kernel Formulation:**

Define the multi-dimensional correlation vector $\mathbf{Z}(\boldsymbol{d}, \boldsymbol{w}) = [\boldsymbol{Z_1}, \boldsymbol{Z_2}, \ldots, \boldsymbol{Z_K}]^{\boldsymbol{T}} \in \mathbb{R}^{\boldsymbol{K}}$, encompassing temporal distance, volumetric deviation, gas-auction bid rank, and bridge-contract nonce sequences. The kernel function is expressed as:

$$S(d, w) = \exp\left(-\frac{1}{2}\|\mathbf{Z}(d, w)\|_2^2\right)$$

For a false pairing, independence and Gaussianity of the feature coordinates are explicit null-model assumptions. Correlated gas, amount and timing features do not satisfy them automatically. The score exp(−‖Z‖²/2) has integral (2π)^(K/2) on R^K; its maximum of one does not normalize it as a density.

A homogeneous Poisson background arrival model, if adopted, needs calibration of its intensity and does not establish the distribution of marked transaction features. The single-pair acceptance region is:

$$\mathcal{B}_\epsilon = \left\{ \mathbf{Z} \in \mathbb{R}^K \;\middle|\; \frac{1}{K}\|\mathbf{Z}\|_2^2 \le \epsilon^2 \right\}$$

4. **Exponential Concentration of Error Bounds:**

Under the additional independent N(0,1) null, the sum of squared coordinates is χ^2_K. For $\delta = 1-\varepsilon^2$ in (0,1), a lower-tail Chernoff calculation matches the acceptance event exactly:

$$\Pr(\|Z\|^2/K \le 1-\delta) \;\le\; \exp[(K/2)(\delta + \log(1-\delta))]$$

Substituting $\delta = 1-\varepsilon^2$ gives the same bound without adding an unrelated upper-tail event:

$$\Pr(\|Z\|^2/K \le \varepsilon^2) \;\le\; \exp[(K/2)(1-\varepsilon^2 + \log(\varepsilon^2))]$$

For fixed ε in (0,1), this expression decays exponentially in K within the stipulated model. Adding correlated, constant or poorly encoded features does not create independent information and may invalidate both the covariance calculation and the bound. Block timestamps also do not supply universally accurate sub-second execution times. The result is therefore a conditional statistical statement rather than a guarantee of entity attribution. ■

Mathematical Definition 1.3 Cross Chain Feature Distance and Candidate Links

Cross-chain linkability is represented by explicit features in a common numeric space. A score between two different event domains does not by itself define the self-distances and reverse distances required for a metric. Any bridge transport and reverse mapping must have stated domains and be compatible with the common representation.

Definition 1.3.1 (Cross-Chain Event Space). Let $\Omega_A \subset C_A$ and $\Omega_B \subset C_B$ denote the compact topological spaces of event logs generated on chains A and B, respectively. An event log $x \in \Omega$ is defined by the 6-tuple:

$$\mathbf{x} = \left[a_{tx}, \tau_{block}, \nu_{token}, \alpha_{id}, \chi_{gas}, \eta_{nonce}\right]^T$$

where a_{tx} is the transaction hash, τ_{block} is the verified timestamp, ν_{token} is the scalar balance, α_{id} is the asset address, χ_{gas} is the gas priority fee, and η_{nonce} is the account execution nonce.

Definition 1.3.2 (Coupling Metric Operator). Let f map events from both chains into a common numeric feature space. For a positive-definite covariance Σ, define the Mahalanobis distance by the square root of the quadratic form:

$$d_C(x_A, x_B) = \sqrt{[(f(x_A) - f(x_B))^T \Sigma^{-1} (f(x_A) - f(x_B))]}$$

A proposed bridge transport Γ may be used when constructing f for chain-B events. The following source feature correspondence is a model specification requiring implementation, not a proved universal deterministic bridge inversion:

$$\mathbf{\Gamma}(\mathbf{x}_B) = \begin{bmatrix} \mathcal{H}_{relayer}(a_{tx,B}) \\ \tau_{block,B} - \mu_{latency} \\ \mathcal{E}^{-1}(\nu_{token,B}) \\ \text{BridgeMapping}(\alpha_{id,B}) \\ \chi_{gas,B} \cdot \text{GasExchangeRate} \\ \eta_{nonce,B} - \Delta_{nonce} \end{bmatrix}$$

Σ is the covariance matrix of the chosen numeric features. Transaction hashes and asset IDs require explicit encodings; constant or dependent features can make Σ singular. Feature selection or specified regularization is then necessary. The square root satisfies the usual metric properties in feature space; the

original squared expression generally violates the triangle inequality. On events, identical feature encodings can make this a pseudometric.

Definition 1.3.3 (Candidate Cross-Chain Link). Two addresses are candidate links when some associated event pair passes a calibrated distance threshold. This defines proximity, not an equivalence class of legal persons:

$$\inf_{\mathbf{x}_u \in \Omega_A(u),\, \mathbf{x}_v \in \Omega_B(v)} d_{\mathcal{C}}(\mathbf{x}_u, \mathbf{x}_v) \leq \gamma_{critical}$$

The critical threshold must be calibrated to its intended error model. If unrelated-pair squared distances follow χ^2_6, accepting below the 1−α quantile accepts 1−α of unrelated pairs. If χ^2 instead models true pairs, a separate background distribution is needed for false positives. Threshold proximity is generally nontransitive. Taking transitive closure creates a new clustering relation that may join distant endpoints and still does not prove common ownership.

1.4 Proposition 1.4 Custodial Exposure and Conditional Identity Attribution

Proposition 1.4. Let G_actor be a candidate cluster of N addresses and let E_off connect some of its vertices to documented VASP deposit addresses. Custodial records can associate a deposit with a customer and support an attribution hypothesis. Such flows do not deterministically identify every sender with that customer. Under explicitly independent edge-success events within each path and independent path-success events across paths, the following reliability product is valid:

$$C_{attr} = 1 - \prod_{k=1}^{M} \left(1 - \prod_{e \in \mathcal{P}_k} P\,(e \in E_{actor}) \right)$$

P_1,…,P_M are supported paths to custodial addresses. Edge-disjointness is a graph property; it does not establish probabilistic independence. The expression must not be treated as a posterior identity probability without priors, likelihoods and ownership evidence.

Proof

1. **Properties of Custodial Sinks:**

A regulated Virtual Asset Service Provider (VASP) maintains an internal relational ledger $\mathcal{L}_{internal}$ decoupled from the underlying blockchain consensus state. When value is transferred via edge $e = (u, v_{deposit})$ where $v_{deposit} \in V_{VASP}$:

$$\Delta_{chain}(v_{deposit}) \Longrightarrow \Delta_{\mathcal{L}}(\text{Account}_{internal})$$

For a specified VASP implementation, a deposit address can receive funds and then be swept into an omnibus account by the custodian. Its behavior and key-control arrangements must be confirmed from records rather than assumed for every custodial service:

$$e_{sweep} = (v_{deposit}, v_{omnibus})$$

In that model the custodian controls the sweep independently of the depositing sender's private key. The address can therefore serve as an observed endpoint for tracing, but its role as a “sink” is an analytical boundary: the funds may continue to move through the VASP's internal or external infrastructure.

2. **Legal Inversion Mapping:**

Custodial compliance records may define a mapping K from deposit addresses to verified customer files. The original references to the BSA and FATF identify relevant legal context, but do not prove that K is injective. The record can contain fields such as the following, subject to actual availability and verification:

$$\mathcal{K}(v_{deposit}) = \mathbf{ID}_{real} = \begin{bmatrix} \text{Legal Name} \\ \text{Tax Identification Number (TIN/SSN)} \\ \text{Physical Domicile Coordinates} \\ \text{Primary Banking Settlement Coordinates (IBAN/ABA)} \end{bmatrix}$$

If each deposit address has one documented customer, K is a well-defined function on that set. Several addresses can still map to the same customer, so uniqueness per address does not imply injectivity or bijectivity. Shared or reassigned deposit infrastructure may require additional account, memo and time information.

3. **Conditional Entity Clustering:**

An equivalence relation used for clustering must be explicitly constructed or independently justified. Its mathematical requirements are retained below; the preceding heuristic thresholds do not automatically satisfy all of them:

- Reflexive: $\forall u \in V_{actor},\ u \sim_E u$.
- Symmetric: $\forall u, v \in V_{actor},\ u \sim_E v \implies v \sim_E u$.
- Transitive: $\forall u, v, w \in V_{actor},\ (u \sim_E v) \wedge (v \sim_E w) \implies u \sim_E w$.

An off-ramping edge shows that value reached a custodial deposit address, with whatever authorization the source protocol requires. It does not imply common ownership of the source and the recipient. The original proposed inference is therefore replaced by the distinction:

$$\text{transfer(u, v) does not imply owner(u)} = \text{owner(v)}$$

If a separately established ownership relation is transitive, it can support propagation only after both relevant ownership links have been proved:

$$\text{owner(w)} = \text{owner(u) and owner(u)} = \text{owner(v) imply owner(w)} = \text{owner(v)}$$

The customer mapping K is defined on documented custodial addresses, not automatically on every address in the cluster. Applying it gives:

$$\mathrm{K}(\mathrm{v}_{\mathrm{deposit}}) = \text{documented customer,} \quad \text{subject to record verification}$$

For example, Alice can pay Bob's VASP deposit address without becoming Bob or sharing ownership with him. The existence of that payment is compatible with the graph and defeats the unconditional identity-collapse conclusion. Extend attribution to other addresses only using independently supported control and identity evidence.

4. **Multi-Path Reliability Product:**

Let $\mathcal{P}_k = \langle e_{k,1}, e_{k,2}, \ldots, e_{k,L_k} \rangle$ be a directed path of length L_k originating within V_{actor} and terminating at a distinct VASP deposit address $v_k \in V_{VASP}$.

The probability that path $\mathcal{P}_k$ is unbroken by external non-attribution errors is given by the product of its independent edge probabilities:

$$P(\mathcal{P}_k) = \prod_{j=1}^{L_k} P\left(e_{k,j} \in E_{actor}\right)$$

If the M path-success events are independent, failure of every path has the product probability below. Edge-disjointness alone does not satisfy this condition: disjoint paths may depend on the same uncertain clustering or identity premise.

$$P(Failure) = \prod_{k=1}^{M}\big(1 - P(\mathcal{P}_k)\big) = \prod_{k=1}^{M}\left(1 - \prod_{e\in\mathcal{P}_k} P\,(e \in E_{actor})\right)$$

Within that reliability model, the probability that at least one path succeeds is one minus the probability that all fail:

$$C_{attr} = 1 - \prod_{k=1}^{M}\left(1 - \prod_{e\in\mathcal{P}_k} P\,(e \in E_{actor})\right)$$

For M ≥ 3 genuinely independent path-success events, each with probability at least 0.98, the numerical example remains:

$$C_{attr} \geq 1 - (1 - 0.98)^3 = 1 - (0.02)^3 = 0.999992$$

The arithmetic is correct under independence. Perfectly correlated paths with marginal success 0.98 instead have union probability 0.98, and dependent edges invalidate within-path products as well. Neither calculation determines a civil or forfeiture evidentiary threshold. That interpretation needs an appropriate legal and statistical model beyond graph reliability. ■

Mathematical Definition 1.4: Custodial Ingestion and Entity Projection Operator

The mapping of abstract on-chain transaction flows into physical legal entities is formalized through linear projection operators and relational database state vectors.

Definition 1.4.1 (Custodial Transfer Vector). Let $V_{actor} \subset V$ denote the attributed cluster and $V_{VASP} \subset V$ denote the target deposit infrastructure. The total exposure vector $\mathbf{x}_{exposure} \in \mathbb{R}^{|V_{VASP}|}$ is defined by:

$$\mathrm{x_{exposure}} = \mathrm{P_{VASP}}^{\mathrm{T}}\,\mathrm{A_{net}}^{\mathrm{T}}\,\mathrm{P_{actor}}\,1$$

P_actor and P_VASP are the original column-selection matrices. For source-row adjacency, transposition is required to collect actor-to-custodian flows; the final all-ones vector sums over actor columns and gives the stated custodial-vector dimension. A_net must represent one asset or fiat-valued flows before summing unlike assets across time. The original untransposed product without the final sum was a matrix, not this exposure vector.

Definition 1.4.2 (Entity Association Operator). Encode customer records as one-hot vectors in a specified finite numeric space. The original outer-product construction then defines a linear map from custodial-address exposures to customer-associated counts, rather than a Hilbert-space proof of legal identity:

$$\mathbf{\Theta}_{KYC} = \sum_{m=1}^{|V_{VASP}|} \mathbf{ID}_{real}^{(m)} \otimes \mathbf{e}_m^T$$

where $\mathbf{e}_m$ is the standard basis vector in $\mathbb{R}^{|V_{VASP}|}$ and $\otimes$ denotes the outer tensor product.

Definition 1.4.3 (Customer Association Vector). Applying the KYC encoding to positive exposure produces the following summary. It can contain multiple nonzero customer coordinates or a count greater than one for a customer with several deposit addresses:

$$\mathbf{\Psi}_{target} = \mathbf{\Theta}_{KYC} \cdot \mathrm{sgn}\big(\mathbf{x}_{exposure}\big)$$

A nonzero projection indicates positive exposure under the encoding; it is not equivalent to any confidence threshold because the expression contains no confidence variable. A confidence-gated decision must

explicitly include a separately calibrated confidence score and threshold. The vector alone does not identify the singular party against whom liability may be assessed.

Phase 2 Cost Basis Assumptions and Conditional Liability Assessment

2.1 Proposition 2.1 The Zero Basis Scenario and Provenance Evidence

Proposition 2.1. Let the provenance record associate an asset disposition with its documented acquisition and subsequent basis adjustments. Accepted basis is an externally established accounting and evidentiary input. A complex graph can make substantiation difficult, but neither cryptographic entropy nor the absence of a contiguous on-chain path proves that accepted basis equals zero. Conditional on a separately justified zero-basis decision, the gain identity makes the full proceeds the modeled gain. The following discussion preserves the original zero-basis scenario while making its status as an assumption explicit.

Proof

1. **Statutory Definition of Basis:**

Basis and gain are separate from graph topology. For a classified property disposition with gross proceeds V_proceeds and accepted basis β_true, signed gain is their difference. Under a stipulated proportional rate τ_cg, the simplified liability expression is:

$$L_k = \left(V_{proceeds} - \beta_{true}\right) \cdot \tau_{cg}$$

The rate, treatment of losses, applicable basis and allocation of the substantiation burden require jurisdiction- and fact-specific rules. They are not determined by this equation. Calling the difference a tax amount also assumes that the rate model applies to that gain and that no additional aggregation or loss provisions alter the result.

2. **Provenance as a Contiguous Path Integral:**

A sequence P = (v_0,…,v_k) can organize acquisition and adjustment evidence from the initial on-ramp to liquidation. A contiguous on-chain path is one possible record structure, not a mathematical necessity for all admissible documentation: external books, transfer references and custodian records can preserve lot information. Let P_admissible denote a calibrated evidentiary probability only if a suitable probability model has been specified.

When a basis history is accepted under the external rules, its accounting identity is:

$$\beta(v_k) = \beta(v_0) + \sum_{j=1}^{k} \Delta\,\beta_{adj}\left(v_{j-1}, v_j\right)$$

3. **Entropy-Induced Path Disconnection:**

The evasion actor deliberately constructs G_{actor} using high-entropy primitives (e.g., decentralized mixers, non-custodial peeling chains, zero-knowledge proofs). Let the cryptographic entropy of the transaction set be S_{crypto}.

One could hypothesize an exponential relationship between a defined entropy statistic and the probability of successful substantiation. The following expression is a model proposal, not a derived legal or statistical law:

$$\mathbb{P}_{admissible}(\mathcal{P} \mid G_{actor}) \propto \exp\left(-c \cdot S_{crypto}\right)$$

Here c, the definition of entropy, the normalizing constant and the calibration data would all need to be supplied. Increasing graph depth does not necessarily drive entropy to infinity, and complete off-chain records can coexist with a complex graph. Even a probability tending to zero would not by itself determine the accepted basis of an individual transaction.

4. **Basis Invalidation and Maximization of Liability:**

A thresholded basis rule can be defined as a scenario assumption. It must not be presented as an established statutory default based solely on graph entropy. If the specified rule independently rejects all claimed basis for a particular disposition, it sets β_accepted to zero; it is the rule and evidence, not the limit of a graph score, that supplies the decision:

$$\beta_{\text{accepted}} = 0 \quad \text{only under the stipulated acceptance decision}$$

Substituting this zero-basis default into the tax liability function yields:

$$L_k = \left(V_{proceeds} - 0\right) \cdot \tau_{cg} = V_{proceeds} \cdot \tau_{cg}$$

Thus the zero-basis identity converts proceeds into modeled gain under the stated premises. It does not prove those premises, maximize liability under all tax systems or show that obfuscation automatically invalidates records. The distinction is essential when reusing the result in later liability sums. ■

Mathematical Definition 2.1: Basis Evaluation Functional

The following threshold functional is a proposed model of basis acceptance. Its output must be distinguished from a legal determination and should be tested against independently substantiated lots before being used as an evidentiary classifier.

Definition 2.1.1 (Provenance Integrity Metric). Define the structural integrity $I\left(v_{target}\right) \in [0, 1]$ of an asset at vertex v_{target} as the product of the inverse out-degrees of all ancestral vertices u in its causal history $\text{Ancestors}\left(v_{target}\right)$:

$$I\left(v_{target}\right) = \prod_{u \in \text{Ancestors}\left(v_{target}\right)} \frac{1}{\max\left(1, \deg_{out}(u)\right)}$$

For D binary ancestors the statistic is exactly 2^(−D); unbranched ancestors contribute one. It is a degree-product score rather than a probability of correct tracing. Identical degree-two graphs can have unequal branch probabilities or complete records. At large D, use exact rationals or logarithms because a floating-point result may underflow to zero even though the mathematical value is positive.

Definition 2.1.2 (Zero-Basis Threshold Functional). Let β_claimed be the asserted basis and I the degree-product score. For this stipulated model, define a candidate accepted-basis output by the following threshold rule; its legal use requires a separate justification:

$$\beta_{accepted} = \mathcal{F}_{basis}(\beta_{claimed}, I) = \beta_{claimed} \cdot \mathbb{I}\left(I\left(v_{target}\right) \geq \epsilon_{audit}\right)$$

Here ε_audit is a supplied model threshold, not a proven statutory quantity. The inequality accepts the full claimed basis at equality and sets it to zero below the threshold, a discontinuous all-or-nothing rule. The script validates this arithmetic and its boundaries, not the legal validity of the rule or the appropriateness of the chosen threshold.

2.2 Lemma 2.2: Atomic Disposition via Intermediate Swap Topology (Graph Traversal Liability)

Lemma 2.2. Let D_swap be a set of uniquely identified dispositions selected by an independently justified classification policy. A single execution can contain multiple transfer edges; changing asset identifiers alone does not establish one taxable event per edge. Given accepted bases, proceeds and rates for these events, the aggregate modeled base liability is their specified sum. Distinct events need not be statistically independent, and independence is not required for additive accounting. Monotonicity and depth growth depend on the gain and loss conventions and the amount of capital in each counted event.

Proof

1. **Property Exchange as Atomic Dispositions:**

The original discussion cites property-exchange rules, including IRC § 1001, as potential legal context. Whether a particular ETH-to-wBTC exchange, bridge movement, wrap/unwrap or internal transfer is a disposition is an external classification question. It cannot be inferred universally from graph traversal or from a change in the token's identifier. The numerical model below stipulates which events are included.

For each intermediate edge $e = (u, v) \in E_{swap}$ occurring at block time t_e, the evader disposes of volume ν_u of asset α_A to acquire volume ν_v of asset α_B.

2. **Fair Market Value (FMV) Oracle Mapping:**

Define a fiat valuation oracle M(asset, quantity, time) returning total proceeds, or equivalently use a unit-price function p(asset,time) multiplied by a quantity in whole-token units. These are different function signatures and must not be mixed. The event-level total-value convention is:

$$V_{fiat}(e) = \mathcal{M}(\alpha_A, \nu_u, t_e)$$

For the original zero-basis scenario, set the accepted basis of each counted disposition to zero as an explicit input. This does not follow from Proposition 2.1 or from a missing metadata field. Assigning zero basis again after every acquisition is an additional assumption: an alternative model giving acquired assets the preceding exchange value can produce different, including negative, signed gains.

3. **Linear Superposition of Intermediate Taxable Events:**

The signed gain of an event is proceeds minus accepted basis. In the separately stipulated positive-gain-only scenario, apply clipping to zero as an additional loss convention. With zero basis and nonnegative proceeds this becomes:

$$g(e) = \max\left(0, V_{fiat}(e) - \beta(u)) = V_{fiat}(e)\right)$$

With each classified disposition counted once and the specified rate applied, the aggregate below is an additive accounting identity. It does not require event independence and does not authorize ignoring loss provisions or resetting acquired-lot basis. The original final-off-ramp comparison is therefore conditional on the chosen event policy:

$$L_{base} = \tau_{cg} \sum_{e \in E_{swap}} g\,(e) = \tau_{cg} \sum_{e \in E_{swap}} \mathcal{M}\,(\alpha_e, \nu_e, t_e)$$

4. **Monotonic Scaling with Obfuscation Depth:**

Let the actor attempt to launder a constant principal fiat value V_0. For a peeling chain that executes N_{swaps} intermediate cross-chain or DEX routing transactions, the fiat value preserved at each hop k is $V_k = V_0 \cdot \left(1 - \phi_{avg}\right)^k$, where $\phi_{avg} \ll 1$ is the average network fee per swap.

The total base liability evaluates as a geometric series:

$$\mathrm{L_{base}} = \tau_{\mathrm{cg}} \mathrm{V_0 q}(1 - \mathrm{q^N})/(1 - \mathrm{q}), \quad \mathrm{q} = 1 - \phi_{\mathrm{avg}}, \quad 0 < \mathrm{q} < 1$$

The exact geometric sum is monotone in N for positive inputs, but its increments diminish and it approaches the finite ceiling τ_cg V_0q/(1−q). The approximation τ_cg V_0N requires cumulative fee-depth effects to be small, with first-order relative error about ϕ_avg(N+1)/2. At zero fee the linear expression is exact; at fee one, post-fee values are zero for k starting at one. Exponential growth in elapsed time under an interest convention must not be confused with growth in the number of hops. ■

Mathematical Definition 2.2: Graph Traversal Tax Functional

The computation of intermediate liability operates as a tensor trace over the unified multi-chain adjacency matrix defined in Phase 1.

Definition 2.2.1 (Disposition Mask). Use an event-level classification label, or a tensor mask Ω_(i,j,α,k), when a source-destination pair can contain mixed swap and non-swap movements. The pair-only matrix Ω_(i,j) is sufficient only when every aggregated cell has one unambiguous disposition interpretation. Unique event identifiers are needed to prevent counting multiple edges of one execution as several gains.

Definition 2.2.2 (Base Liability Contraction). Let M_(α,k) be fiat price per whole token, not an already-total fiat valuation. For quantity tensor A in corresponding token units and an unambiguous mask, the zero-basis model is:

$$L_{base} = \tau_{cg} \sum_{i,j,\alpha,k} A_{i,j,\alpha,k} \Omega_{i,j,\alpha,k} M_{\alpha,k}$$

This contraction reproduces a sum of classified zero-basis event proceeds under the units and aggregation assumptions. An epoch price must be constant or appropriately quantity-weighted to reproduce individual event valuations. Multiplying quantities by an oracle output already expressed as total proceeds would count quantity twice. The tensor operation establishes an accounting result, not a tax classification or an automatic increase in liability for every ledger hop.

2.3 Theorem 2.3 The Insolvency Paradox as a Conditional Crossing Model

Theorem 2.3 (Conditional Insolvency Threshold). Let L(N) be modeled debt after N full-principal zero-basis events, and let modeled wealth be $V_0 q^N$ plus off-chain assets W_off. For fixed retention q in (0,1) and fixed positive multiplier m incorporating the chosen rate, penalty and common accrual factor, define $C = mV_0 q/(1-q)$. A finite depth at which debt exceeds this equality wealth model exists if and only if C > W_off. Without that condition, no universal critical depth follows. If the wealth expression is only an upper bound, exceeding it is sufficient but not necessary for insolvency.

Proof

1. **Civil Fraud Multiplier (The Aggravating Factor):**

The original civil-fraud example uses a penalty rate of 75%. A complicated address network does not by itself prove fraud or show that the entire underpayment is eligible. Let L_eligible denote the portion independently determined to satisfy the penalty rule and ρ_fraud the applicable modeled rate. The general adjusted principal is:

The penalized principal tax liability becomes:

$$L_{fraud} = L_{base} + \rho_{fraud} L_{eligible}$$

Only when all modeled base liability is penalty-eligible does this reduce to L_base(1+ρ_fraud). Substituting the exact geometric sum under that additional premise gives:

$$L_{fraud} = \tau_{cg}(1 + \rho_{fraud}) V_0 q(1 - q^N)/(1 - q)$$

2. **Continuous Temporal Compounding:**

Continuous compounding is one mathematical convention for an already-existing liability with no new additions during the interval. The annualized rate r(t), day-count convention, accrual start dates and treatment of penalties must be specified externally. A transaction date is not automatically the legal start date for tax or penalty interest.

For a component satisfying dL/dt = r(t)L between its start date and the evaluation date, the solution is the exponential integral below. Different components may need different starting dates or separate penalty and tax factors:

$$L_{total}(t_{now}) = L_{fraud} \cdot \exp\left(\int_{t_0}^{t_{now}} r(t)\, dt \right)$$

For a constant average interest rate $\bar{r}$, this simplifies to:

$$L_{total}(N, \Delta t) = \tau_{cg}(1 + \rho_{fraud}) e^{r \Delta t} V_0 q(1 - q^N)/(1 - q)$$

3. **Decay of Net Asset Value (Network Friction):**

A simplified model assumes routed capital decays by a constant fractional friction per full-principal event. Actual account balances can also change through prices, splits, merges, new contributions and external movements. Consequently, applying this recurrence to the whole portfolio requires evidence that the event sequence and units satisfy that model.

Assuming an average fractional friction coefficient $\phi \in (0, 1)$ per hop, the terminal net worth strictly from the evaded funds is:

$$V_{net}(N) \leq V_0(1-\phi)^N + V_{off_chain}$$

where V_{off_chain} represents legally acquired off-chain assets (e.g., real estate, bank accounts). Let the total global initial wealth be $W_0 = V_0 + V_{off_chain}$.

4. **Solving for the Critical Topological Depth (N_{crit}):**

The Insolvency Paradox occurs when the legally assessed debt exceeds the evader's total global wealth:

$$L_{total}(N) > V_{net}(N)$$

Substituting the derivations:

$$\mathrm{C}(1-\mathrm{q}^{\mathrm{N}}) \;>\; \mathrm{V}_0\mathrm{q}^{\mathrm{N}} + \mathrm{W}_{\mathrm{off}}, \quad \mathrm{C} = \mathrm{m}\mathrm{V}_0\mathrm{q}/(1-\mathrm{q})$$

Write m = τ_cg(1+ρ_fraud)exp(rΔt) only within the common-factor scenario. The exact debt is C(1−q^N), not a line growing without bound, while routed principal is V_0q^N. Debt therefore approaches C and modeled total wealth approaches W_off. If C ≤ W_off, they do not cross at any finite depth in the equality model. This corrects the extrapolation of the small-fee linear approximation to arbitrary N.

When C > W_off, algebra gives the following strict crossing condition; the critical integer is the smallest admissible N satisfying it:

$$\mathrm{q}^{\mathrm{N}} \;<\; (\mathrm{C} - \mathrm{W}_{\mathrm{off}})/(\mathrm{C} + \mathrm{V}_0)$$

For example, V_0 = 100, q = 0.9 and m = 0.35 give C = 315. Off-chain wealth of 400 prevents a crossing at every depth. With q = 0.999, C = 34,965 remains below off-chain wealth of 40,000 despite the small per-hop fee. In the separate zero-fee model, N > (V_0+W_off)/(mV_0) gives a strict crossing. None of these calculations establishes legal liability or the completeness of a real balance sheet. ■

Mathematical Definition 2.3: Total Liability Tensor Contraction

The liability timeline can be expressed as a vector contraction once each event's accepted basis, applicable rate, penalty eligibility and accrual start date are supplied. The scalar output is a modeled amount for analysis, not mathematical authorization for legal filing or immediate collection.

Definition 2.3.1 (Punitive Accrual Vector). Define the temporal liability vector $\mathbf{L}_{base} \in \mathbb{R}^K$ where $\mathbf{L}_{base,k}$ represents the base tax generated during epoch k from the tensor trace in Definition 2.2.2.

Let b_k be base liability and e_k its penalty-eligible portion. If one factor applies to each adjusted component, define an upper-triangular accrual matrix using its specified start date; where tax and penalty accrual differ, use separate vectors or factors. The original continuous model is:

$$\mathbf{C}_{k,j} = \begin{cases} \exp\left(\int_{t_k}^{t_j} r\,(t)\,dt\right) & \text{if } j \geq k \\ 0 & \text{if } j < k \end{cases}$$

Definition 2.3.2 (Final Modeled Demand). At the evaluation date T, sum each base component and its eligible penalty with the appropriate accrual factor. The resulting scalar is a modeled demand amount and has no embedded authorization predicate:

$$L_{demand}(T) = \sum_k (b_k + \rho_k e_k) C_{k,T}$$

The original simplified contraction (1+ρ_fraud)L_base^{T}C_:T is recovered only when eligibility and accrual conventions permit it. A 75% penalty on 40 of a base 100 produces 130, not 175. Continuous and daily compounding are different conventions. Neither the resulting demand nor the number of graph hops establishes fraud, a jeopardy condition or authority to bypass procedural requirements.

Phase 3 Flight Risk Enforcement Models and Constrained Recovery

3.1 Proposition 3.1 Capital Flight Stopping Time and Intervention Conditions

Proposition 3.1. Let F_t denote telemetry available up to time t, and let liquid value follow the stipulated killed-drift process below. Define flight as either its first killing jump or depletion below a fixed positive threshold. A longer delay can reduce expected available value, but finite delay and positive rates alone do not imply near-zero recovery. Real-time observation cannot guarantee that intervention precedes an unpredictable jump. The model describes available value under its assumptions, not a proof that any named privacy protocol or offshore custodian makes funds mathematically untraceable.

Proof

1. **Capital Flight as a Stochastic Process. Let V_liquid(t) be value remaining within the modeled recoverable domain. The equation below describes deterministic proportional depletion and a killing jump. It contains no Brownian term and is therefore not a jump-diffusion model:**

$$dV_{liquid}(t) = -\mu(t)V_{liquid}(t)\,dt - V_{liquid}(t^-)\,dN(t)$$

Here μ(t) is nonnegative drift and N_t supplies the first killing event with deterministic locally integrable intensity λ(t). A stochastic intensity needs its own conditional model and averaging. For first jump T_J, the solution is $V_0 \exp(-\int_0^t \mu(s)ds)\ 1[T_J > t]$. Economic interpretations of a killing event or changes after notice require independent calibration.

2. **The Stopping Time and Recovery Probability: The flight stopping time τ_{flight} is formally defined as:**

$$\tau_{flight} = \inf\{t \geq 0 \mid V_{liquid}(t) \leq \epsilon\}$$

Take a fixed threshold $0 < \varepsilon < V_0$ and let t_ε be the first time the no-jump path reaches it. Then τ_flight = min(T_J,t_ε). With constant positive drift, $t_\varepsilon = \log(V_0/\varepsilon)/\mu$. Any procedural delay Δt_audit is an external input; the original 30–90-day example and its legal relevance are not proved by the process. The corrected survival probability is:

$$\Pr(\tau_{flight} > t) = 1[t < t_\varepsilon]\exp(-\int_0^t \lambda(s)ds)$$

The indicator supplies the deterministic threshold cutoff omitted from the original equation. Before that cutoff, expected available value is $V_0 \exp(-\int_0^t (\mu+\lambda))$. Asymptotic decay to zero requires divergent cumulative depletion/hazard or threshold crossing. Positive integrable rates can instead leave a positive limiting balance. A divergence of λ after notice is an additional assumption, not a mathematical consequence of sending notice.

3. **Intervention Timing. The discussion of jeopardy procedures identifies a possible legal mechanism whose availability, conditions and timing need separate verification. In the abstract model, reducing delay can improve expected available value. Let an actually feasible and authorized intervention occur at t_seize; its modeled available value before flight is:**

$$V_{rec} = V_{liquid}(t_{seize}) \cdot \mathbb{I}\left(t_{seize} < \tau_{flight}\right)$$

For a fixed positive reaction delay Δ and constant hazard λ, the probability that the jump occurs first is 1−exp(−λΔ). Observing survival now does not predict the next jump. A zero-delay intervention with positive initial value is an idealized boundary case, not a telemetry guarantee. Execution, access and authorization also remain prerequisites to realized recovery. ■

Mathematical Definition 3.1: Jeopardy Trigger Functional

A continuous risk score can organize telemetry and liability inputs for decision support. It cannot mathematically supply the legal elements needed to authorize emergency action.

Definition 3.1.1 (Flight Risk Tensor). Define the behavioral risk vector $\mathbf{r}(t) \in \mathbb{R}^3$, composed of the first derivative of transaction velocity $v'(t)$, the liquidity ratio of the entity matrix, and a heuristic interaction proximity score to known mixer contracts.

Definition 3.1.2 (Dimensionless Risk Threshold). For L_demand > 0, choose a positive reference amount L_ref in the same monetary units and define a proposed indicator on finite, normalized risk features:

$$\Gamma_{\mathrm{J}}(\mathrm{r}, \mathrm{L}) = 1[(\mathrm{w}^{\mathrm{T}}\mathrm{r})\log(\mathrm{L}/\mathrm{L}_{\mathrm{ref}}) \geq \gamma]$$

The ratio replaces the unit-dependent logarithm of a monetary amount. Dollars-to-cents conversion then leaves the score unchanged if L_ref is converted too. Specify the feature units, weights, signs and calibration of γ. A negative weighted score can make the trigger decrease with liability, so monotonicity is not automatic. Equality activates the displayed weak-threshold rule. Its output is not a statutory finding or permission to act.

3.2 Lemma 3.2: Induced Cryptographic Illiquidity via Access-Control State Mutation

Lemma 3.2. Consider a particular stablecoin contract with a specified privileged blacklist mechanism and transfer guard. Setting the relevant flag blocks successful transfers that consult that guard while the flag remains set. It does not necessarily make the state absorbing, freeze unrelated assets, eliminate transaction attempts or prove legal ownership. A privileged administrative reset, upgrade or alternative execution path must be considered in the actual implementation. The Boolean model below proves the conditional restriction, not permanent universal illiquidity.

Proof

1. Standard State Transition of Token Transfers. Let S contain a balance mapping M_bal and the relevant authorization state. A request to transfer quantity ν from u to v succeeds only if its complete predicate P(S,u,v,ν) holds, including the applicable balance, signature or contract-authorization conditions and other protocol checks. A valid key or signature alone is insufficient to establish that all these conditions hold.
2. **The Blacklist Precondition. For the modeled implementation, let B(u) and B(v) indicate sender and recipient restrictions. The conjunctive guard is:**

$$P^*(S, u, v) = P(S, u, v) \wedge \neg\mathcal{M}_{blacklist}(u) \wedge \neg\mathcal{M}_{blacklist}(v)$$

3. **Administrative Freeze Operator. If an authorized administrator successfully invokes the appropriate entrypoint under the actual contract and external legal requirements, the modeled mutation sets the specified flags:**

$$\delta_{admin}\left(S, \mathcal{B}_{freeze}(u)\right) \Longrightarrow \mathcal{M}_{blacklist}(u) := 1 \quad \forall u \in A_{evader}$$

The target cannot reset the flag through the modeled ordinary transfer path. Whether administrators can reset it, upgrade the contract or execute another balance operation depends on the implementation. Consensus enforces the implemented rules; it does not make those rules permanently unchangeable.

4. **Conditional Transfer Restriction. While the sender's flag remains one and the modeled guard is evaluated, the conjunction fails. Thus the particular transfer is rejected even if the sender can otherwise authorize it:**

$$P^*(S', u, v) = 0$$

The rejected transfer rolls back under the applicable execution semantics. For successful transfers of the affected token that use this guard, the resulting outgoing capacity is zero during the restriction:

$$\text{outgoing}_c\text{apacity}_{\text{token}}(u, t) = 0 \quad \text{while the stated guard and flag apply}$$

The on-chain balance can remain at u even though these transfers are blocked. The model does not transfer it into the treasury or show that the restriction is irreversible. It also does not establish ownership from the balance mapping. Freeze, seizure, liquidation and receipt are distinct states in any recovery accounting. ■

Mathematical Definition 3.2: EVM Blacklist State Operator

The imposition of illiquidity is modeled as a localized restriction on the adjacency tensor defined in Phase 1.

Definition 3.2.1 (Restricted Adjacency Tensor). Let $\mathbf{A}(t) \in \mathbb{R}^{N\times N\times A}$ be the instantaneous adjacency tensor for asset class $\alpha \in A$ (specifically stablecoins). The execution of the administrative freeze by authority Authdefines a diagonal restriction matrix $\mathbf{R}_{freeze} \in \{0,1\}^{N\times N}$, where:

$$\left(\mathbf{R}_{freeze}\right)_{i,i} = \begin{cases} 0 & \text{if } v_i \in A_{evader} \\ 1 & \text{otherwise} \end{cases}$$

Definition 3.2.2 (Two-Endpoint Capacity Restriction). For source-row prospective adjacency of the affected asset α, both the sender and recipient restrictions require:

$$A_{\text{constrained}}{}^{(\alpha)}(t) = R_{\text{freeze}}\, A_{\text{proposed}}{}^{(\alpha)}(t)\, R_{\text{freeze}}$$

Left multiplication alone zeros frozen source rows but permits transfers into a frozen receiver, contrary to the predicate above. Apply the two-sided mask only to the relevant asset slice and current policy. The Boolean diagonal mask is idempotent, preserves nonnegativity and cannot increase entries, but it does not erase past transfers, enforce all balance constraints or prove that flags persist forever.

3.3 Theorem 3.3 Conditions for Total Recovery in Multilateral Optimization

Theorem 3.3 (Constrained Multilateral Recovery). Let L_demand be an externally established debt input and let the asset set contain uniquely identified on-chain and off-chain holdings with specified ownership, quantities and realization values. Recovery is bounded by debt and realizable accessible wealth. Equality with min(L_demand,W_total) holds only in an ideal model where all included assets are accessible, divisible, realizable at their stated values and subject to no other binding restriction. Coordination can enlarge the feasible set without guaranteeing total financial exhaustion.

Proof

1. **The Recovery Objective. Let x_i be the feasible recoverable fraction of asset i, w_i its quantity, p_i its unit value, and h_i a stipulated realization factor accounting for the model's conversion from value to proceeds. Then the coalition's simplified objective is:**

$$\text{maximize } R = \sum_i x_i w_i p_i h_i, \quad 0 \le x_i \le 1$$

The feasible set must additionally represent jurisdiction, ownership, access, encumbrances, costs, indivisibility and permitted dispositions. Asset IDs must be unique across jurisdiction lists. Within a model imposing a debt cap, recovery satisfies:

$$\mathcal{R} \leq L_{demand}$$

It is also bounded by the value that can actually be obtained from the specified holdings. Gross asset value, net worth and realizable proceeds should not be interchanged. The following wealth bound presumes W_total is defined consistently with that feasible recovery model:

$$\mathcal{R} \leq W_{total}$$

2. **Multilateral Jurisdiction Mapping. The customer associations in Proposition 1.4 can guide further investigation, but they do not automatically identify all legally owned off-chain holdings. A separately documented asset inventory could include the original illustrative categories:**

$$\mathcal{O}_{off} = \{\text{Bank Accounts}_{US}, \text{Real Estate}_{UK}, \text{Brokerage}_{EU}\}$$

Without coordination, an otherwise identical feasible set may be limited to holdings accessible within one jurisdiction. Cooperation, if legally and operationally available, can expand that set and cannot lower the optimum. The existence of a customer record or a treaty name does not prove access to every bank account, property or brokerage asset, nor guarantee that local and global recoveries differ by a particular amount.

3. **Conditions for Attaining the Upper Bound. If the independently established inputs happen to satisfy the inequality below, debt is larger than the wealth measure. Phase 2 does not guarantee that inequality for every complex network:**

$$L_{demand} > W_{total}$$

The two scalar bounds then have the following minimum:

$$\min(L_{demand}, W_{total}) = W_{total}$$

This minimum is only an upper bound unless feasibility and attainability have been proved. In the ideal all-accessible, divisible, full-value model described above, one obtains:

$$\mathcal{R}^* = W_{total}$$

The equality is conditional, not a consequence of the bounds alone. A feasible example with debt 300 and total asset value 200 but an inaccessible holding has maximum recovery 140. Two indivisible assets worth 60 each with a cap of 100 yield 60 in a binary selection model; selling and returning surplus is a different model. Realization discounts can also reduce proceeds. Actual collection requires the corresponding legal and operational conditions. ■

Mathematical Definition 3.3: Multilateral Seizure Operator

A treasury accounting operator must distinguish a restriction or lien from transfer, liquidation and receipt. The following asset representation permits quantity-aware valuation, while actual cash receipts require a cash-flow record.

Definition 3.3.1 (Global Asset Vector). Define the comprehensive asset vector $\mathbf{w} \in \mathbb{R}^M$, composed of Mdiscrete assets (both digital and physical) mapped to the evader via $\mathbf{ID}_{real}$. Let $\mathbf{v}_{val} \in \mathbb{R}^M$be the corresponding current market valuation vector. Total net worth is $W_{total} = \mathbf{w}^T\mathbf{v}_{val}$.

Definition 3.3.2 (Recovery Selection Matrix). Let P_seize(t) be explicitly diagonal with entries in {0,1} for whole-asset selection. Being a general Boolean matrix is not sufficient to make it a projection. Fractional entries can instead model divisible recovery, with different algebraic properties. Entries represent realized recovery only if the selected state is actual receipt, not merely a freeze or lien.

Definition 3.3.3 (State Treasury Ingestion). The cumulative value recovered by the tax authority coalition $\mathcal{R}(t)$is governed by the trace:

$$R(t) = \text{Tr}[P_{\text{seize}}(t)\text{diag}(w \odot v_{\text{val}} \odot h)]$$

Including quantities corrects the original trace, which used unit prices without w. For w = (2,3) and unit prices (10,20), total value is 80, whereas the omitted-quantity trace is 30. The debt cap and feasibility constraints must still be imposed. A constant P(t) = diag(1,0) need never reach identity despite debt exceeding wealth. Assuming every entry eventually becomes one assumes attainability rather than proving it; changing market values also must not be confused with cumulative treasury receipts.

Phase 4 Illustrative Simulation and Forensic Ledger Mathematics

4.1 Proposition 4.1 Mapping the Illustrative Multi Chain Trajectory

Proposition 4.1. The illustrative multi-chain graph spans Ethereum, Arbitrum and Avalanche and contains seven counted event rows after the initial acquisition. Under the supplied zero-basis classifications, 20% rate, 75% penalty eligibility and table-specific interest convention, the arithmetic produces a debt exceeding the terminal routed liquidity. This example does not prove that any network with ten swaps becomes insolvent. Event values, traversed capital, retained holdings and complete wealth are essential inputs to such a claim.

Proof

1. Initial Condition and Basis Assumption. Let the initial principal be V_0. The example labels its acquisition records as missing and assigns zero accepted basis to the subsequently counted events. That assignment is a scenario input, not a consequence established from the routing table. Classification as a taxable disposition and treatment of undocumented basis require separately verified rules and evidence.
2. Graph Traversal Liability Matrix. The source scenario includes peeling, bridge operations and AMM exchanges, with protocols such as Hop and Uniswap mentioned as examples. Each row is treated as one classified event for the calculation; these labels are not proof that every transfer or bridge is taxable. Preserve quantities, asset identities, times and unique event references before applying the event-sum operator, so that several transfer edges from one execution cannot be counted repeatedly.
3. Penalty and Interest Inputs. The source example assumes that all base tax is eligible for a 75% fraud penalty. It does not establish willfulness from a network topology. The cited legal provisions, including any limitation-period consequences, need independent verification rather than inference from the arithmetic. A general model with accepted basis, event-specific rates and accrual dates can use the following expression; its interest factor must match the table convention if the table is being reproduced.

Model convention for the general component formula

$$L_k = V_{fiat,k} \cdot \tau_{cg} \cdot \left(1 + \rho_{fraud}\right) \cdot \exp(r\Delta t)$$

The illustrative rates are τ_cg = 0.20 and ρ_fraud = 0.75. Continuous exponential interest is a possible mathematical model, but the displayed simulation table instead uses an additional 5% of base tax. These conventions are distinguished below rather than silently equated.

4. Terminal Comparison. The retained routed value is lower than the initial principal, and the sum of the stipulated event liabilities exceeds that terminal value. At fixed event values and fixed accrual factors, accumulation is linear in event count; under positive constant fractional fees it is a bounded geometric sum. Exponential interest growth in elapsed time is a separate model. The simulation therefore illustrates an assumed accounting scenario rather than inevitable global insolvency. ■

Mathematical Definition 4.1: Ledger Evaluation Operator

The multi-chain trajectory is mathematically evaluated without geometric graphs by organizing the telemetry into a discrete transactional matrix.

Definition 4.1.1 (Typed Simulation Ledger). Retain the original table's nine displayed fields: event index, origin ledger, destination ledger, action and asset transition, gross FMV, assumed basis, modeled base gain,

penalty plus interest, and cumulative debt. The chain and action fields are categorical, so the unencoded table is not an element of R^(N×9). A fuller machine-readable record additionally needs separate consumed/emitted assets and quantities, unique disposition IDs, timestamps, price sources, base tax, penalty eligibility and accrual dates. Those fields were named in the original proposed schema but were not actually present in the displayed table.

Definition 4.1.2 (Cumulative Debt Functional). The total sovereign debt functionally maps the transactional trace to the final Jeopardy Assessment value:

$$L_{demand}(t_{now}) = \sum_{k=1}^{N_{hops}} \left[\left(\mathrm{FMV}_k - \mathcal{F}_{basis}(\mathrm{Claimed}_k, 0) \right) \cdot \tau_{cg} \cdot 1.75 \cdot C_{interest}(t_{now} - t_k) \right]$$

For the displayed scenario, accepted basis is stipulated as zero on each counted row. For a general analysis, supply the accepted basis and event-specific interest factor rather than fixing F_basis to zero from topology. Enforce unique event identifiers: duplicated ingestion would otherwise increase the sum. A disposition of 100 with accepted basis 80 produces an adjusted amount of 7 under the example's 20% rate and 75% penalty before interest, demonstrating the dependence on this input.

4.2 Simulated Multi-Chain Traversal Matrix

The original matrix is retained below. It describes an initial 5,000 ETH position valued at approximately $15,000,000, followed by seven representative events across three chains over a stated twelve-month period. The numerical assumptions are 20% tax, a 75% penalty on all modeled base tax and interest equal to 5% of base tax. The last assumption reproduces one year of simple interest on base tax only; actual event dates and legally applicable accrual rules are not supplied.

Original simulation table with stipulated event classifications and zero accepted basis

Reading the table. “Base Gain” is the modeled gain, not base tax; base tax is 20% of that column. “Penalty & Interest” equals 80% of base tax, and cumulative debt includes base tax as well. The inferred route is 0 → 1 → 2, then branches 3 → 5 and 4 → 6, followed by a merge into 7. The rows are not seven sequential transformations of the entire principal. In particular, the increase from row 4 to row 5 reflects different branches rather than new capital.

Hop	Ledger C_A	Ledger C_B	Action & Asset Transition	Gross FMV ($)	Basis ($)	Base Gain ($)	Penalty & Interest ($)	Cumulative Debt ($)
0	Off-Chain	Ethereum	Initial Acquisition (Missing Records)	15,000,000	0.00	0	0	0
1	Ethereum	Ethereum	DEX Swap: 5,000 ETH →14.9M USDC	14,900,000	0.00	14,900,000	2,384,000	5,364,000
2	Ethereum	Arbitrum	Bridge Lock: 14.9M USDC →Arbitrum	14,850,000	0.00	14,850,000	2,376,000	10,710,000
3	Arbitrum	Arbitrum	Peeling Split: 10M USDC →wBTC	10,000,000	0.00	10,000,000	1,600,000	14,310,000
4	Arbitrum	Arbitrum	Peeling Split: 4.8M USDC →wETH	4,800,000	0.00	4,800,000	768,000	16,038,000
5	Arbitrum	Avalanche	Bridge Burn/Mint: wBTC →AVAX	9,900,000	0.00	9,900,000	1,584,000	19,602,000
6	Arbitrum	Avalanche	Bridge Burn/Mint: wETH →AVAX	4,750,000	0.00	4,750,000	760,000	21,312,000

Hop	Ledger C_A	Ledger C_B	Action & Asset Transition	Gross FMV ($)	Basis ($)	Base Gain ($)	Penalty & Interest ($)	Cumulative Debt ($)
7	Avalanche	Off-Ramp	VASP Consolidation & Liquidation	14,600,000	0.00	14,600,000	2,336,000	26,568,000

4.3 Theorem 4.3 Comparative Balance Sheet and Liquidity Shortfall

Theorem 4.3 (Terminal Liquidity Shortfall). Under the table's stipulated event and assessment inputs, the modeled demand is $26,568,000 and terminal routed liquidity is $14,600,000. Their difference is −$11,968,000. This proves the subtraction and a shortfall against that liquidity pool. It does not establish the actor's complete net worth or recoverability, and the magnitude of the deficit is smaller than the remaining liquidity rather than greater than it.

Proof

1. **Final Routed Value. The source starts with $15,000,000 and ends with $14,600,000 at the VASP consolidation and liquidation row. The implied branch bookkeeping reconciles a $400,000 reduction. Without actual quantities, prices, transaction hashes, timestamps and fees, that fiat-value change cannot be attributed uniquely to network charges rather than price movement or omitted flows. The terminal figure used in the arithmetic is:**

$$V_{liquid}(t_{end}) = 14{,}600{,}000$$

2. **Aggregate Modeled Gains. With zero accepted basis and the supplied disposition labels, the seven counted FMVs sum as follows. Distinct event rows do not imply statistical independence, which is unnecessary for this arithmetic:**

$$\text{Total Gross Realized Gains} = \sum_{k=1}^{7} V_{fiat,k} = 73{,}800{,}000$$

The 20% modeled rate produces base tax of $14,760,000. The stipulated 75% penalty adds $11,070,000. The table's interest is exactly $738,000, equal to 5% of base tax, making total debt $26,568,000. One year of continuous 5% interest on the penalized principal would instead use $25,830,000 × exp(0.05). It fits the source total only at another assumed common age, about 0.5634 years; the missing accrual dates do not establish that age.

Reconciled total under the table-specific interest convention

$$L_{demand} = 26{,}568{,}000$$

3. **Terminal Liquidity Deficit. Subtract the modeled demand from the terminal routed liquidity:**

$$\mathcal{D}_{final} = V_{liquid}(t_{end}) - L_{demand}$$

$$\mathcal{D}_{final} = 14{,}600{,}000 - 26{,}568{,}000 = -11{,}968{,}000$$

The resulting $11,968,000 shortfall is not a complete balance sheet. Additional assets of $20,000,000 and other liabilities of $1,000,000 give modeled total net worth of $7,032,000 after this debt. With no other liabilities, $11,968,000 of additional assets brings net worth to zero. Under the inferred branch bookkeeping and no other assets, cumulative debt first exceeds routed portfolio value at row 4, not at a universally fixed depth. Ten equally valued full-principal events at the stated tax and penalty rates would yield debt of 3.5 times one event value, whereas ten small events need not exhaust a separately retained holding. No assessment or recovery authority follows from these comparisons. ■

Phase 5: Cryptographic Primitives and Multi-Chain Verification Mathematics

5.1 Proposition 5.1: Non-Linear State Authorization via ECDSA over secp256k1

Proposition 5.1. For a protocol using ECDSA over secp256k1, a signature can satisfy the algebraic verification relation for a message and public key. Signature validity is one authorization condition, not a sufficient condition for transaction validity on every ledger: balances, nonces, encoding and other state rules still apply. The equation does not identify a unique legal person, establish sole key possession or prove intent. The detailed curve and signature derivation below is retained within that technical scope.

Proof

1. **Domain Parameters and Key Generation:**

The secp256k1 curve is defined over the finite field $\mathbb{F}_p$ by the Weierstrass equation:

$$y^2 \equiv x^3 + 7 \pmod{p}$$

where $p = 2^{256} - 2^{32} - 2^9 - 2^8 - 2^7 - 2^6 - 2^4 - 1$.

Let G be the universally defined base point (generator) with prime order n. The evader generates a private key $d_A \in [1, n-1]$ and computes the public key locus $Q_A = d_A \cdot G$.

2. **Signature Generation:**

For a correctly defined transaction-message hash converted to integer z under the protocol's rules, choose a suitable secret nonce k in {1,...,n−1}. Compute the curve point kG. Signing must reject any zero r or s and follow the required encoding and nonce-generation rules. The signature components are:

$$r \equiv x_1 \pmod{n}$$

The signature proof s is mathematically bound to the private key d_A via the congruence:

$$s \equiv k^{-1}(z + r \cdot d_A) \pmod{n}$$

3. **Consensus Verification and Limits of Attribution**

A verifier first checks the signature ranges 1 ≤ r,s < n, validates the public point and required group membership, and applies the specified hash and encoding rules. It then computes:

$$u_1 \equiv z \cdot s^{-1} \pmod{n}, \quad u_2 \equiv r \cdot s^{-1} \pmod{n}$$

The network recovers the point $(x_2, y_2) = u_1 \cdot G + u_2 \cdot Q_A$.

By substituting $Q_A = d_A \cdot G$ and s:

$$u_1 \cdot G + u_2 \cdot Q_A = (z \cdot s^{-1})G + (r \cdot s^{-1})(d_A \cdot G) = s^{-1}(z + r \cdot d_A)G = k \cdot G$$

A valid signature must produce a point other than infinity with x-coordinate congruent to r modulo n. The substitution proves the algebra for a correctly generated signature. Selected public keys and a signature agree with an independent cryptography implementation, but tests do not prove computational hardness. Ordinary ECDSA also admits (r,n−s) when (r,s) is valid; additional protocol rules may reject one form. A valid signature can accompany an overspend or wrong nonce, and shared, delegated or compromised keys defeat an inference to one legal identity or intent. The conclusion concerns the signature relation, not universal ledger validity or legal non-repudiation. ■

5.2 Lemma 5.2: Hierarchical Deterministic Key Derivation (BIP-32) and Structural Provenance

Lemma 5.2. Under BIP32, a seed generates an extended private key, comprising a private scalar and chain code, and a corresponding extended public key. For non-hardened indices $i < 2^{31}$, public derivation from the parent public point, chain code and index agrees with private derivation followed by multiplication by G.

This relation permits traversal of specified normal descendants, not arbitrary hardened branches or automatic identification of every used address. The original derivation is retained with those scope conditions.

Proof

1. **Parent Key Extension:**

An extended private key is formally represented as the tuple (k_{par}, c_{par}), where k_{par} is the normal 256-bit private key and c_{par} is the 256-bit chain code functioning as structural entropy. The corresponding extended public key is (K_{par}, c_{par}), where $K_{par} = k_{par} \cdot G$.

2. **The Child Key Derivation (CKD) Function:**

To derive a normal (non-hardened) child key at index i, the algorithm computes a 512-bit intermediate hash sequence I:

$$I = \text{HMAC-SHA512}\left(c_{par}, \text{ser}_P\left(K_{par}\right) \parallel \text{ser}_{32}(i)\right)$$

The resulting hash sequence is bisected into two 32-byte vectors: $I = I_L \parallel I_R$.

The child private key k_i is algebraically defined over modulo n:

$$k_i \equiv \left(\text{parse256}(I_L) + k_{par}\right) \pmod{n}$$

The returned chain code c_i for the child is simply assigned the right-half sequence I_R.

3. **Public Key Derivation and Forensics Mapping:**

Due to the distributive property of elliptic curve cryptography, multiplying the private key derivation equation by the generator G yields:

$$k_i \cdot G = \left(\text{parse256}(I_L) + k_{par}\right) \cdot G$$
$$K_i = \text{parse256}(I_L) \cdot G + k_{par} \cdot G$$
$$K_i = \text{parse256}(I_L) \cdot G + K_{par}$$

Normal child derivation uses both the parent public point and chain code; the point alone is insufficient. The public/private relation above matches selected BIP32 standard vectors and several normal indices. An extended public key cannot cross a hardened derivation edge, whose input requires private material. Invalid left scalars, zero private children or public points at infinity require the specified rejection and index-advance policy. These checks do not establish full wallet conformance. Index ranges, hardened boundaries, address encodings, ledger activity and ownership evidence are also required before using derivation relations for attribution. ■

5.3 Theorem 5.3 Cryptographic State Commitments and Merkle Patricia Trie Verification

Theorem 5.3. Let σ_t be the Ethereum world-state mapping at a specified block under its finality assumptions. A correctly constructed Merkle Patricia Trie root is a computational commitment to the encoded state. It does not ensure strict synchrony across other Layer-1 or Layer-2 systems, encode legal identity or contain every historical event. The state transition and commitment equations below are useful technical relations, but their evidentiary interpretation requires authenticated roots, correct proof types and the actual bridge protocol.

Proof

1. **World State Formalization:**

World State Formalization. The state maps account addresses to their account fields. Construct state-trie key-value pairs using the specified address-key derivation and account serialization, including nonce, balance, storage root and code hash. A sequence of account payloads alone is not enough to build the trie because their keys and path encoding are also required. The source extraction notation is retained as shorthand for that complete keyed representation:

$$L_S(\sigma) \equiv \{p(a) : \sigma[a] \neq \emptyset\}$$

where $p(a)$ represents the Recursive Length Prefix (RLP) encoded payload of the account parameters.

2. **Merkle Patricia Trie (MPT) Construction:**

MPT Construction. The modified radix trie uses branch, extension and leaf nodes with nibble paths and compact-path flags. Node contents are RLP encoded. Child references may be hashes or embedded short nodes according to the encoding rules; they are not always hashes. The root commits to this specific encoded structure under collision-resistance assumptions.

Let TRIE(I) denote the root node hash of the trie structure constructed from the item set I. The block header H permanently records this root hash as the state root H_r:

$$\mathrm{TRIE}\big(L_S(\sigma)\big) = H_r$$

3. **Cross-Chain Relayer Proof Validation:**

Cross-Chain Proof Validation. If a destination protocol uses source-chain membership proofs, it must authenticate the relevant source root and apply its own finality and event-processing policy. Not every bridge uses the same light-client mechanism. State, transaction and receipt roots commit to different objects; choose the root and proof appropriate to the claimed fact.

An Ethereum MPT proof supplies the encoded nodes needed along the requested key's path. Verification decodes each node, checks its reference, follows the correct nibble or branch position, and confirms the terminal value. A simple list of binary-tree sibling hashes is not a sufficient MPT specification.

The validation operation is therefore represented as a path-aware predicate, rather than the original repeated concatenation formula:

$$\mathrm{VerifyMPT}(\mathrm{root}, \mathrm{key}, \mathrm{value}, \mathrm{encoded_nodes}) = \text{true or false}$$

Successful membership against a supplied root is only one condition for any destination mint or release. Root authentication, finality, the correct event semantics, replay protection and the destination's execution constraints must also hold. A valid proof can be rejected because the root is untrusted or the event has already been processed.

4. **Forensic Handoff Integrity:**

Forensic Handoff Integrity. The tested fixtures cover inline and hashed child references and reject wrong paths, values, missing leaves and modified nodes. Changing a committed balance changes the fixture root, demonstrating sensitivity without proving mathematical impossibility of collisions. These fixtures do not cover all malformed wire data, non-membership proofs, full client construction or live headers. A finalized source event may be delivered later to its destination, so a commitment does not prove strict cross-chain synchrony. Multiple events can also share a block timestamp. Identity correlation, clock accuracy and legal significance remain independent issues. ■

Phase 6 Reconciliation of the Forensic Ledger Matrix

6.1 Theorem 6.1 Liability and Asset Decay under the Stated Inputs

Theorem 6.1. The supplied Phase 6 figures reuse the Phase 4 event values. Under the same zero-basis, rate, penalty and interest assumptions, they reproduce the debt and terminal liquidity shortfall exactly. This is arithmetic reconciliation of one illustrative ledger, not a second empirical observation or a proof of total forfeiture. A debt-to-remaining-principal ratio can grow geometrically while absolute debt remains bounded, so these quantities must be distinguished.

Proof

1. **Asset Decay Under Bridge Verification Costs:**

   The routing scenario spans Ethereum, Arbitrum and Avalanche. A simplified per-step recurrence can subtract AMM and gas fractions from the same incoming value, provided units match and the fractions are nonnegative with sum at most one:

   $$v_{k+1} = v_k \cdot (1 - \phi_{AMM}) - \phi_{gas}$$

   Given V_0 = \$15,000,000 and terminal routed value \$14,600,000, the exact implied reduction is $2/75$, approximately 2.6667%. The source approximation 0.026 would give \$14,610,000 if used exactly, a \$10,000 difference. Cumulative friction is one minus the product of per-step retention factors, not generally the sum of fees. Sequential charges use different bases: two 10% charges leave 81 from 100, whereas two charges on the same initial base leave 80. Fixed gas amounts and reserve-dependent execution need further modeling.

2. **Recursive Generation of Taxable Mass:**

   A cryptographic record can support the occurrence of an event without determining its tax classification or accepted basis. The calculation below retains the source assumption that all seven selected rows are distinct zero-basis dispositions. Because the table branches, its listed values are not sequential whole-portfolio balances and cannot validate a seven-step full-principal fee recurrence merely by fitting the endpoints.

   The total realized gain G_{total} is the arithmetic sum of the fair market value at each distinct hop:

   $$G_{total} = \sum_{k=1}^{N_{hops}} V_k = 73{,}800{,}000 \text{ USD}$$

3. **Application of Penalty Tensor:**

   Apply the source scenario's scalar inputs, a 20% gain rate and a 75% penalty on all modeled base tax. These reproduce the accounting identities without establishing their legal applicability:

   $$L_{base} = 0.20 \times 73{,}800{,}000 = 14{,}760{,}000 \text{ USD}$$

$$L_{punitive} = L_{base} \times (1 + 0.75) = 25{,}830{,}000 \text{ USD}$$

The final source demand is $26,568,000. Its implied interest of $738,000 equals 5% of base tax only. It must not be described as established continuous compounding over the unspecified event timeline. The original table, event dates and accepted accrual convention would have to agree before making that interpretation.

4. **Terminal Exhaustion Deficit:**

The final state deficit $\mathcal{D}_{final}$ evaluates to:

$$\mathcal{D}_{final} = V_{terminal} - L_{demand} = 14{,}600{,}000 - 26{,}568{,}000 = -11{,}968{,}000 \text{ USD}$$

This deficit measures debt relative to the routed terminal liquidity, not complete net worth. In the constant-retention model, L(N)/(V_0q^N) = [mq/(1−q)](q^(−N)−1), while L(N) approaches mV_0q/(1−q). For V_0 = 100, q = 0.99 and m = 0.35, debt first exceeds remaining routed principal at N = 3, but its ceiling 3,465 stays below off-chain wealth of 4,000. Zero basis alone does not specify rates, event values, count or wealth. The figures therefore do not necessitate immediate liquidation or prove access to other assets; recovery remains subject to Phase 3's feasible set. ■

Phase 7 Graph Correspondence and Corporate Tax Architecture Analogies

7.1 Proposition 7.1: Topological Homomorphism Between Peeling Chains and Circular Intercompany Transfer Pricing

Proposition 7.1. A peeling-chain graph G_actor and a corporate transaction graph G_corp can be compared through an explicit mapping of vertices and edges. A graph homomorphism exists only if the map preserves every source edge in the target graph. It can merge vertices and does not imply graph isomorphism, equality of economic attributes or legal equivalence under IRC § 482. The corporate example below is retained as an analogy whose formal validity must be checked on specified graphs and weights.

Proof

1. **The Corporate Evasion Graph (Transfer Pricing):**

The Corporate Graph. The original paper invokes IRC § 482 and an arm's-length transfer-pricing framework as legal context. Its scope and application to actual controlled taxpayers require independent analysis. For the mathematical analogy, let vertices c_1,…,c_n represent subsidiaries and an edge represent a transfer of goods, intangibles or services at a stated price P_tp. Control relationships and the difference between that price and a benchmark are additional attributes, not facts proved by an edge.

A stylized base-erosion scenario routes profits across jurisdictions with different modeled rates. This describes one possible optimization problem; it does not establish that every multinational routing pattern is evasive or that all corporate edges correspond to blockchain transfers.

2. **Homomorphic Mapping of Nodes and Edges:**

Define a proposed vertex map Φ from V_actor to V_corp and require (u,v) ∈ E_actor to imply (Φ(u),Φ(v)) ∈ E_corp. This edge-preservation requirement must be checked; merely naming analogues does not establish existence for arbitrary graphs.

- **Vertices:** A pseudonymous intermediate wallet $u \in V_{actor}$ maps to an intermediary shell corporation or conduit entity $c_i \in V_{corp}$ residing in a low-tax jurisdiction.

- **Edges:** A blockchain state transition $e_{crypto} = (u, v)$ with volume v and network fee ϕ maps to an intercompany transaction $e_{corp} = (c_i, c_j)$ executed at transfer price P_{tp}.

The payment/change split can be compared conceptually with a corporate transaction that allocates receipts between an operating entity and another group entity. In both examples one can track selected and residual amounts. The comparison does not establish that a payment price is artificial, that a recipient is a tax haven conduit, or that the relevant monetary weights are preserved by Φ. These details must be represented explicitly in any claimed equivalence.

3. **Artificial Allocation and the Arm's Length Standard:**

For the stylized transfer-pricing comparison, let P_arm be a supplied benchmark arm's-length price under an independently justified method. The mathematics can compare P_tp with that benchmark, while the applicable statutory standard and legal consequences remain external premises.

The artificial divergence in corporate evasion is $\Delta P = P_{tp} - P_{arm}$.

In the cryptographic model (Lemma 1.2), the divergence is measured by the mantissa entropy of the change output $D_{KL}(\mathbf{p}(o_c) \parallel \mathbf{u})$.

The price deviation and the mantissa divergence are different quantities: one can be currency per unit, while KL divergence is dimensionless. No transformation preserving them or their legal meaning has been established. A fork-to-edge homomorphism can merge vertices without being an isomorphism, and a directed path need not map into a single loop-free directed edge. A cyclic corporate graph also cannot be isomorphic to Phase 1's event DAG. Heuristic clustering therefore does not prove statutory reallocation of income or collapse the actual corporate graph. ■

Mathematical Definition 7.1: Transfer Pricing Elasticity Tensor

Definition 7.1.1 (Transfer Pricing Shift Matrix). Let $\mathbf{T}_{corp} \in \mathbb{R}^{n \times n}$ be the matrix of intercompany transaction volumes. The profit shifting vector $\mathbf{s} \in \mathbb{R}^n$ representing the capital drained via base erosion is defined by:

$$\mathbf{s} = \left(\mathbf{T}_{corp} \odot \mathbf{E}_{tp}\right) \cdot \mathbf{1}$$

The expression computes row sums of B = T_corp ⊙ E_tp. Define units: quantity times a currency-per-unit deviation yields currency, while dimensionless deviations require a currency-valued T for that interpretation. Zeroing E_tp makes the resulting vector zero but leaves T_corp and its edges intact. Conversely, signed deviations can cancel to zero. For a conservative transfer-flow adjustment, incoming minus outgoing is $B^T1 - B1$ and sums to zero; the displayed outgoing row totals need not. None of these operations establishes a legal adjustment or graph isomorphism.

7.2 Lemma 7.2: Economic Substance Doctrine and Graph Contraction Operators

Lemma 7.2. A selected path P = (v_0,…,v_k) may be simplified by explicitly eliminating intermediate vertices while preserving a specified property such as reachability. Whether a transaction lacks economic substance is a separate question and is not equivalent to graph contraction. Zero holding duration or hedged exposure does not prove zero benefit. The following utility and contraction discussion is retained as a conditional model, with the invalid matrix-power limit replaced by explicit elimination.

Proof

1. **Economic Substance in Corporate Law:**

Economic Substance in Corporate Law. The source invokes economic-substance and step-transaction concepts to motivate the comparison. Their complete legal tests and application require independent review. A mathematical utility function can represent chosen economic inputs, but no scalar sign or graph property is substituted here for those legal elements.

2. **Evaluating On-Chain Economic Utility:**

Define the economic utility function of an intermediate cryptographic hop $U(v_i)$. For a legitimate financial transaction (e.g., liquidity provisioning, yield farming), $U(v_i)$ incorporates expected yield $\mathbb{E}[Y]$, delta exposure Δ, and duration δ_t.

For a stylized automated flash-swap or bridge path, holding duration at an intermediate vertex may be small and some market exposure may be hedged or atomic. Other benefits, including a service or change in access to liquidity, can remain even when duration tends to zero. A cost-only model must explicitly set all such benefits to zero.

Under that additional zero-benefit assumption, utility is minus the specified costs. It is strictly negative only when their sum is positive; zero costs give zero utility:

$$\mathrm{U}(\mathrm{v_i}) = -\phi_{\mathrm{gas}}(\mathrm{v_i}) - \phi_{\mathrm{slippage}}(\mathrm{v_i}) \leq 0$$

3. **The Step Transaction Graph Contraction:**

The negative value follows from the assumed utility model. It does not independently demonstrate intent, eliminate unmodeled benefits or establish a legal violation. An instantaneous service with benefit 10 and costs 5 gives positive utility in a valid countermodel.

Graph Simplification. The source analogy with the Step Transaction Doctrine can motivate examination of an entire path. In mathematics, eliminating intermediate vertices is an explicitly defined representation change; any change to legal event classification or deductions needs a separate rule.

Define the contraction operator $\mathcal{K}_{step}$ on the path $\mathcal{P}$:

$$\mathcal{K}_{step}(\langle v_0, v_1, \dots, v_k \rangle) \Longrightarrow e_{direct} = (v_0, v_k)$$

For a simple path, an explicit shortcut can connect v_0 and v_k. If used only for visualization, keep the original disposition records for liability accounting. If an external rule instead replaces several dispositions by one endpoint event, recompute gain and basis under that rule. The earlier event sum and an endpoint-only assessment generally differ and cannot both be substituted without reconciliation. ■

Mathematical Definition 7.2: Step Transaction Contraction Functional

Definition 7.2.1 (Substance Indicator Function). For an intermediate vertex $v \in V_{actor}$, define the substance indicator $\chi_{sub}(v) \in \{0, 1\}$ based on the evaluation of the utility vector $\mathbf{u}_v = [\delta_t, \mathbb{E}[Y], \mathbf{Vol}(v)]^T$:

$$\chi_{sub}(v) = \mathbb{I}\left(\mathbf{w}_{sub}^T \mathbf{u}_v > \tau_{substance}\right)$$

Here the threshold and weights belong to a proposed classifier, not a proven legal standard. Normalize duration, yield and volatility, or transform the weights when units change. At equality the strict greater-than rule returns zero. The choice of weights, calibration data, benefit terms and threshold must be specified before interpreting the result.

Definition 7.2.2 (Explicit Graph Contraction). Partition the adjacency into retained vertices R and eliminated vertices E. If the eliminated subgraph is a DAG containing m vertices, its nilpotent adjacency permits the finite shortcut construction:

$$\mathrm{A^*_{RR}} = \mathrm{A_{RR}} + \mathrm{A_{RE}}(\mathrm{I} + \mathrm{A_{EE}} + \cdots + \mathrm{A_{EE}}^{\mathrm{m-1}})\mathrm{A_{ER}}$$

Use Boolean operations for reachability and then remove E. Numerical path products and sums need a separately stated weight interpretation and do not automatically conserve money. For $0 \to 1 \to 2$, eliminating 1 adds the $0 \to 2$ shortcut. The original limit of powers of A diag(1−χ) is zero on a DAG and can oscillate or diverge on cyclic graphs; it does not create this edge. Right masking also retains the columns with $\chi = 0$ rather than deleting those vertices. An undefined A_valid cannot supply a proof unless its shortcut construction is given independently.

7.3 Theorem 7.3 Conditional Routing Correspondence and Treaty Shopping Analogies

Theorem 7.3 (Conditional Routing Correspondence). Cross-chain routing and a stylized corporate conduit network may share a path-selection formulation. A true optimization equivalence requires a bijection of feasible paths preserving the relevant costs and constraints. Neither their similar descriptions nor their objective syntax establishes that property. The retained treaty-shopping analogy does not prove that cross-chain movements force an effective tax rate to zero or are legally equivalent to a specific corporate structure.

Proof

1. **The Corporate BEPS Routing Matrix:**

The Corporate Routing Example. The source's "Double Irish with a Dutch Sandwich" illustration describes a conceptual route from a high-tax jurisdiction through a Netherlands conduit to a low-tax destination such as Bermuda. It is retained as historical context for a stylized graph, without asserting the current availability or tax effect of that arrangement. Assign explicit modeled rates to edges before comparing routes.

Let W_tax contain proportional loss or withholding-rate inputs for the stylized model. If each rate applies to value remaining after the preceding edge, a path retains the product of its factors (1−W_tax). Minimizing loss requires maximizing that product, or minimizing the sum of negative logarithms, rather than minimizing retention.

2. **The Cryptographic Bridge Routing Matrix:**

In the cryptographic analogy, a path may be evaluated for asset accessibility, contract restrictions and reporting exposure. Those attributes must be defined and supported. An analyst's ability to infer a link is distinct from a contract's ability to transfer a token.

Let C_A and C_B be two ledgers connected through a specified bridge B. Whether a direct route exists and what administrative restrictions or compliance checks apply are protocol-specific questions. The bridge can be represented as an intermediate vertex for comparison with a conduit, but it does not thereby inherit the legal or economic function of a corporate holding company.

Let $\mathbf{C}_{risk} \in \mathbb{R}^{L\times L}$ be the matrix of cross-chain regulatory exposure across L ledgers.

3. **Equivalence of the Optimization Functions:**

The corrected proportional-retention objective for the stylized corporate graph is:

$$\text{maximize}_{\text{P}} \prod_{(\text{i,j})\in \text{P}} (1 - \text{W}_{\text{tax}}(\text{i}, \text{j}))$$

If C_risk is interpreted as adverse-event probabilities, and independence or valid conditional survival factors justify multiplication, the corresponding survival objective is:

$$\text{maximize}_{\text{P}} \prod_{(\text{A,B})\in \text{P}} (1 - \text{C}_{\text{risk}}(\text{A}, \text{B}))$$

For factors in (0,1], maximizing their product is equivalent to minimizing the sum of −log(factor); a rate of one needs separate infinite-cost handling. Define allowed paths and cycle rules. Minimizing retention, as in the original formula, selects larger losses and can favor endless lossy cycles. Marginal probabilities alone also do not justify a survival product: two perfectly shared events with survival 0.9 still have joint survival 0.9, not 0.81. If C_risk is merely a score, it has no probability interpretation without calibration. A cost-preserving path map must be exhibited to claim equivalence.

The source's comparison with a global minimum tax is therefore conceptual, not an algebraic identity. An edge-by-edge 15% floor differs from an aggregate 15% floor: two 15% proportional losses compound to 27.75%. This does not state how any actual policy is implemented. Likewise, computing a cross-chain attribution metric changes information available to the analyst; it does not automatically overwrite physical restrictions, enforce a tax floor or make two networks isomorphic. ■

Phase 8 Cost Basis Records and Conditional Valuation Algorithms

8.1 Lemma 8.1 Wallet Lot Records and Basis Preservation Conditions

Lemma 8.1 (Lot Records Across Wallets). A portfolio can be represented as separate wallet-level sets of acquisition lots. Transfers can preserve quantity, basis and acquisition time through associated records even when these fields are not stored on-chain. Address fragmentation does not force the probability of successful identification to zero. FIFO selects by acquisition time, not by highest taxable gain, and unknown basis is not automatically zero. The source's wallet-by-wallet legal discussion must therefore be separated from the following bookkeeping model.

Proof

1. **The End of Universal Pool Accounting:**

The original manuscript cites IRC § 1012(c), Treasury Regulation § 1.1012-1(j) and a January 1, 2025 transition as its regulatory premise. The validation scripts do not establish the accuracy, scope or effective application of those claims. Treat the applicable lot-identification and account-allocation rules as external inputs requiring independent verification. Separate wallet accounting can be modeled without assuming that historical basis disappears when an asset moves between wallets.

2. **Wallet-Localized Basis Vectors:**

Define the cost basis state of a specific wallet $w \in W$ at block time t as a vector of unspent lots:

$$\mathbf{B}_w(t) = \langle (v_1, \beta_1, t_{acq,1}), (v_2, \beta_2, t_{acq,2}), \ldots, (v_k, \beta_k, t_{acq,k}) \rangle$$

where v_i is the token volume, β_i is the localized fiat cost basis, and $t_{acq,i}$ is the acquisition timestamp.

For a homogeneous lot with quantity Q and total basis B, a stipulated proportional transfer of q units moves basis Bq/Q and leaves B(1−q/Q), preserving total basis when no separate adjustment applies. The acquisition time can remain associated with the transferred lot. Whether the transfer is a taxable disposition, and which allocation method is acceptable, are external classification questions rather than consequences of this vector identity.

3. **Information Loss in Obfuscation Graphs:**

An on-chain transfer may omit acquisition metadata, but an external ledger keyed by the transfer reference can preserve it. Generating single-use addresses therefore does not logically strip the complete evidence record. In a genuinely incomplete record, represent missing basis and date explicitly rather than substituting numeric values:

The resulting basis vector in the receiving wallet v initializes with empty provenance:

$$\mathbf{B}_v(t) = \langle (v_{in}, \emptyset, \emptyset) \rangle$$

4. **Unknown Basis and FIFO Allocation**

Under a stipulated FIFO policy, consume available lots from earliest acquisition time, with a defined tie rule and checks on quantity. Other identification policies may require different evidence. A missing on-chain record alone does not prove that the owner lacks all contemporaneous records or that any particular legal default must apply.

FIFO does not convert absent acquisition cost into zero. The validation allocator rejects unknown basis until a separately justified value or rule is supplied. Conditional on zero accepted basis from such an external decision, the original gain identity is retained:

$$G_{taxable} = \text{Proceeds} - 0 = \text{Proceeds}$$

FIFO is not universally gain-maximizing. Selling for 120 using an older basis-100 lot gives gain 20; choosing a newer basis-10 lot would give gain 110. The ordering can produce the opposite comparison when the older lot has lower basis. Partial lots also require proportional accounting: consuming three units from a

total-basis-90 three-unit lot and one unit from a total-basis-100 two-unit lot allocates basis 140. These calculations clarify the algorithm without establishing any jurisdiction's required method. ■

8.2 Theorem 8.2: Continuous Pricing Oracle Evaluation via Time-Weighted Average Price (TWAP) Integrals

Theorem 8.2. A TWAP is the time integral of a defined price process over a positive observation window divided by its duration. It can damp short-lived deviations under specified amplitude and duration bounds. The integral does not by itself establish legally accepted FMV, immunity to manipulation or economic prohibitivity of affecting an average. Pool selection, quote direction, observation quality, liquidity and applicable valuation rules remain explicit inputs to any assessment.

Proof

1. **The Vulnerability of Discrete Spot Pricing:**

The Vulnerability of Spot Observations. A pool's momentary price can differ from other market observations or be affected by a large trade. A single spot reading is therefore not necessarily representative of a chosen valuation window. This motivates comparing time-weighted observations, without proving that one trade can eliminate a legally assessed gain or that TWAP always supplies the required legal valuation.

2. **The TWAP Accumulator Algorithm:**

AMM designs such as the Uniswap v2 mechanism considered in validation can maintain cumulative price variables. These support time averaging when the observations and update semantics are correctly interpreted; the accumulator alone does not neutralize every source of price distortion.

Let $P(t)$ be the spot price of the asset at continuous time t. The accumulator state variable $C(T)$ stores the integral of the price history from the contract's inception t_0 to block timestamp T:

$$C(T) = \int_{t_0}^{T} P\,(t)\,dt$$

For a price held constant between updates, the cumulative increment is the previous interval's price multiplied by elapsed time. In the referenced Uniswap v2 design, updates occur when the relevant contract routine executes, using elapsed time and prior reserves; they are not autonomous actions at the beginning of every block. The recurrence is:

$$C(T_n) = C(T_{n-1}) + P(T_{n-1}) \cdot (T_n - T_{n-1})$$

3. **Derivation of the Bounded Average:**

To calculate a TWAP over a positive-duration interval, obtain compatible cumulative observations at both endpoints, including a justified extrapolation if necessary. The result is the difference of cumulative readings divided by elapsed time:

$$P_{TWAP} = \frac{C(T_{end}) - C(T_{start})}{T_{end} - T_{start}}$$

A perturbation need not persist for the entire averaging window to change the result. A price of 70 for 60 seconds in an otherwise 100-valued 1,800-second signal gives TWAP 99. More generally, if deviation magnitude is at most M over duration d in a window T, the average changes by at most Md/T. Averaging damps a bounded short deviation but does not establish immunity or its economic cost.

4. **Tax Assessment Application:**

Let a modeled disposition occur at t_e and transfer whole-token quantity Q_e. Multiplying by a fiat-per-token TWAP gives modeled proceeds under those units. A centered 30-minute window needs observations until 15 minutes after execution and is therefore a later calculation, not an immediately known price. The source valuation formula is retained with that availability condition:

$$V_{proceeds} = \nu \cdot P_{TWAP}(t_k - 15, t_k + 15)$$

The cumulative identity is exact for a step-held process; for a continuously changing signal a left-price recurrence is generally an approximation. Unequal durations require time weights, partial intervals must be integrated consistently, zero-duration updates contribute nothing, and a zero or reversed window is invalid. Inverting a TWAP is generally different from averaging reciprocal prices, so quote direction must also be fixed. These mathematical properties do not decide legally accepted FMV or guarantee that a reference pool represents the wider market. ■

8.3 Proposition 8.3: Constant-Product Invariant Mechanics in Intermediate Realization

Proposition 8.3. In a constant-product model with positive reserves, the effective-input invariant yields a definite output quantity for a specified input and fee. Multiplying the output by a consistent fiat quote gives modeled proceeds, and subtracting accepted basis gives signed gain. The bonding curve itself has no variables establishing tax classification, holding-period treatment or accepted basis, so it does not force a legally binding realization conclusion independently of those premises.

Proof

1. **The AMM Invariant State:**

Let a decentralized exchange pool hold reserves of Asset X (e.g., USDC) and Asset Y (e.g., ETH). The pool is governed by the constant-product invariant:

$$R_X \cdot R_Y = k$$

Here x and y are positive reserve quantities. Without retained fees the stated invariant describes the usual idealized trade relation. With a retained fee, distinguish effective fee-adjusted reserves from actual balances; the actual product can increase during a trade.

2. **The Disposition Mapping:**

To obfuscate the trail, the evader deposits an input volume Δx of Asset X into the pool to extract an output volume Δy of Asset Y. Accounting for the protocol liquidity fee ϕ (e.g., 0.3%), the effective input is $\Delta x_{eff} = \Delta x \cdot (1 - \phi)$.

For Δx ≥ 0 and fee f in [0,1], effective input is Δx(1−f). The effective-reserve transition satisfies:

$$\left(R_X + \Delta x_{eff}\right)(R_Y - \Delta y) = R_X \cdot R_Y$$

Solving for the output volume Δy:

$$\Delta y = \frac{R_Y \cdot \Delta x_{eff}}{R_X + \Delta x_{eff}}$$

3. **Tax Realization via Asset Transition:**

The swap records an exchange between asset quantities in the AMM model. Its tax treatment and any change in holding period require separate rules. As corrected in Lemma 8.1, missing on-chain acquisition metadata does not establish zero accepted basis. Set β to the accepted-basis input for the specific classified disposition rather than inferring β = 0 from the pool mechanics.

Using a consistent fiat-per-output-token quote, signed modeled gain is:

$$G = \Delta y \cdot P_{TWAP} - \beta$$

Substitution of the effective-input output function gives:

$$G = [y\Delta x(1-f)/(x+\Delta x(1-f))]P_{TWAP} - \beta$$

Exact rational fixtures verify the output identity and $0 \le \Delta y < y$ for admissible inputs. Positive reserves, input and price with $f < 1$ give positive proceeds; gain is positive only if accepted basis is smaller than proceeds. Actual reserves receive the full input when the fee remains in the pool, so $(x+\Delta x)(y-\Delta y)$ exceeds xy for positive input and fee. The finite execution ratio $\Delta y/\Delta x$ also differs from marginal price y/x. Rounding output down in atomic units introduces less than one unit of error and makes the effective invariant an inequality. Convert atomic quantities before fiat valuation: 1,500,000 units with six decimals at 2 fiat per token is 3 fiat. None of these identities establishes tax liability without the external premises. ■

Phase 9: Mathematical Algorithms in Enforcement and Asset Seizure

9.1 Proposition 9.1: Deterministic Storage Slot Resolution and Merkle Trie Mutation in Smart-Contract Freezing

Proposition 9.1. ***Let $\sigma_{contract}$ denote the storage state tree of a centralized stablecoin contract (e.g., ERC-20 USDC or USDT) hosted on an Ethereum Virtual Machine (EVM) ledger. Let $p \in [0, 2^{256}-1]$ be the declared storage slot position of the administrative blacklist mapping:***

$$\text{mapping(address} \Longrightarrow \text{bool)} \quad \text{blacklisted}$$

For an address u and a mapping base slot p established from the actual deployed layout, the storage position $\kappa(u,p)$ is computed from fixed-width encodings and Keccak-256. A successful authorized write can change the designated flag and storage commitment. Transfer blocking follows only for execution paths that consult that flag. The write is not inherently irreversible: another authorized write can restore the previous value. Collision resistance is a computational security assumption, not injectivity over every possible input.

Proof

1. **EVM Storage Layout. Model storage as a key-value map over 256-bit words. Solidity storage allocation can involve packing and inheritance, and proxy or custom layouts require additional resolution. The base slot cannot always be obtained by simply counting declarations. For a mapping with an address key, concatenate the key left-padded to 32 bytes with the base slot encoded as a 32-byte unsigned big-endian integer:**

$$\kappa(u,p) = \text{Keccak256}\left(\text{pad}_{256}(u) \parallel \text{pad}_{256}(p)\right)$$

2. **Storage Root Mutation. The account state commits to a storage root built under the protocol's trie encoding rules. In this example, the designated flag is initially zero. If the actual contract exposes a permitted blacklist entrypoint and the external authorization conditions are satisfied, an administrator may submit the modeled transaction:**

$$T_{freeze} = \text{freezeAddress}(u)$$

For the assumed Boolean mapping and entrypoint, successful execution performs the relevant storage write. The isolated operator represents the following value update without claiming to emulate all contract effects:

$$\mathcal{S}_{t+1}[\kappa(u,p)] := 1$$

This modification cascades up the branches of the storage trie:

$$\rho_{storage}^{(t+1)} = \text{UpdateStorageTrie}\left(\rho_{storage}^{(t)}, \kappa(u,p), \text{RLP}(1)\right)$$

which subsequently mutates the global world state root:

$$H_r^{(t+1)} = \text{UpdateWorldState}\left(\sigma_t, a_{contract}, \rho_{storage}^{(t+1)}\right)$$

3. **Execution Reversion Under the Specified Guard. A transfer path that actually includes the stated blacklist check evaluates the following condition. Such a guard is not part of every token implementation or necessarily every entrypoint:**

$$\text{SLOAD}\big(\kappa(u,p)\big) == 0$$

During EVM bytecode execution of $T_{transfer}$, the engine evaluates:

$$\text{condition} = \text{SLOAD}\big(\kappa(u,p)\big)$$

When this guard observes a set flag, the modeled transfer reverts. An ordinary REVERT rolls back changes in the failing frame while preserving unused gas, subject to its execution costs; it does not necessarily consume the entire gas limit. If an administrator is permitted to reset the flag, later transfers can become valid. A restored isolated storage root does not imply restoration of the global state root, since nonce, gas and other state may differ. Thus conditional restriction, permanence and treasury recovery are separate properties. ■

Mathematical Definition 9.1: Storage Space Pre-Image Mapping

The storage calculation uses fixed-length bit-string encodings and a finite key-value state map. Calling the entire storage space a real vector space or invoking a binary field does not establish the semantics of an implemented contract operation.

Definition 9.1.1 (Fixed-Width Preimage Encoding). For $0 \le u < 2^{160}$ and $0 \le p < 2^{256}$, define a 512-bit preimage by exactly padding the address and slot to 256 bits each:

$$E_{slot}(u, p) = 0^{96} \parallel \text{bin}_{160}(u) \parallel \text{bin}_{256}(p)$$

Here bin_d means a fixed-width unsigned big-endian bit string, and ‖ denotes concatenation. The result is exactly 64 bytes, including at p = 0. The original $256-\log_2(p)$ padding was undefined at zero, nonintegral for most p, and one bit too long for powers of two when combined with their ordinary binary digits.

Definition 9.1.2 (State Isolation Functional). For a storage map S over 256-bit words, the isolated flag-setting operator leaves every other modeled entry unchanged and sets the selected entry to one:

$$\left(\Phi_{freeze}(\mathcal{S})\right)_k = \begin{cases} 1 & \text{if } k = \text{Keccak256}\big(\mathcal{E}_{slot}(u,p)\big) \\ \mathcal{S}_k & \text{otherwise} \end{cases}$$

The definition models one storage mutation. It does not establish administrative authority, guard coverage, immutable code or permanent absence of outgoing flows. The selected slot must match the actual layout, and the transaction must be valid and successfully executed. The validation uses small offline trie fixtures, not a production EVM or live contract invocation.

9.2 Lemma 9.2: (t, n)-Threshold Cryptographic Custody via Shamir Polynomial Reconstruction for Multi-Agency Forfeiture

Lemma 9.2. Let a custodial private scalar be shared among n participants through a t-of-n Shamir scheme over a prime field, with $1 \le t \le n < q$ and distinct nonzero coordinates. Uniform independent masking coefficients make fewer than t shares reveal no additional information about the secret under the shares-only model, preserving its prior. Any t valid authentic shares reconstruct the secret. The no-single-holder property requires $t \ge 2$. Sharing itself does not implement distributed key generation, threshold signing, policy control or safeguards against misuse by an authorized quorum.

Proof

1. **Polynomial Construction over Prime Field $\mathbb{F}_q$: Let $q$be the large prime defining the group order of the secp256k1 elliptic curve:**

$$q = 2^{256} - 432420386565659656852420866394968145599$$

Corrected domain parameter. The standard group order is hexadecimal FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141, distinct from the coordinate-field prime in section 5.1. The original subtraction constant 432420386565659656852420874635263103247 produced a value 8,240,294,957,648 smaller than this order. That source value is composite: the script computes 2^(q_old−1) mod q_old ≠ 1. A prime-field proof cannot use that modulus. The reconstruction validations explicitly use the corrected standard order.

Choose the secret constant term a_0 = d_seize and sample all t−1 other coefficients independently and uniformly from F_q. The resulting random polynomial has degree at most t−1. If the secret is a curve private scalar, restrict it to the nonzero scalar domain and state that prior separately:

$$f(x) = d_{seize} + \sum_{l=1}^{t-1} a_l\, x^l \pmod{q}$$

Each masking coefficient is sampled from the entire field, including zero. Forcing the leading coefficient to be nonzero changes the distribution. In the exhaustive small-field fixture, that exact-degree condition can exclude a candidate secret after fewer-than-threshold observations and therefore changes the privacy argument.

2. **Share Distribution: Each agency $j_m \in J$is assigned a public evaluation coordinate $x_m \in \mathbb{F}_q \setminus \{0\}$, where $x_m \neq x_k$for $m \neq k$. The private share issued to agency $j_m$is:**

$$y_m = f(x_m) \pmod{q}$$

3. **Information-Theoretic Security against $k < t$Collusion: Assume a sub-coalition of $k$agencies $\mathcal{C}_{sub} \subset J$colludes to recover d_{seize}, where $|\mathcal{C}_{sub}| = k < t$. The colluding agencies possess the system of linear congruences:**

The share equations have rank k for distinct nonzero coordinates. Including the secret among the unknowns leaves t−k degrees of freedom. Once a candidate secret is fixed, there are q^(t−1−k) compatible coefficient tuples of degree at most t−1, not one in general. Uniqueness holds only at k = t−1. Uniform masks give equal likelihood of the observed shares under each secret, so the correct posterior statement is:

$$\Pr(\text{secret} = s \mid \text{fewer than t shares}) = \Pr(\text{secret} = s)$$

For a uniform secret on the full field, this posterior equals 1/q; a nonuniform or nonzero-scalar prior is preserved instead of becoming uniform. The statement conditions only on shares. Public keys and other auxiliary information require their own model. Share privacy does not itself establish inability to execute every unauthorized transfer or prove the legal control of a custody system.

4. **Deterministic Reconstruction via Lagrange Interpolation: Let a quorum of $t$authorized agencies $Q \subseteq J$assemble their shares $\{(x_i, y_i)\}_{i=1}^{t}$. The secret $d_{seize} = f(0)$is computed using Lagrange basis polynomials evaluated at $x = 0$:**

$$\ell_i(0) = \prod_{\substack{j=1 \\ j \neq i}}^{t} \frac{0 - x_j}{x_i - x_j} = \prod_{\substack{j=1 \\ j \neq i}}^{t} \frac{-x_j}{x_i - x_j} \pmod{q}$$

The recovered master private key is:

$$d_{seize} = \sum_{i=1}^{t} y_i \, \ell_i(0) \pmod{q}$$

A threshold-signature protocol may produce signatures without reconstructing the key in one process, but this requires a complete, separately specified protocol. FROST is threshold Schnorr signing, not Threshold-ECDSA; the scheme must be compatible with the destination. Distributed key generation, authenticated messages, verifiable shares, availability and policy enforcement are additional requirements. The tested interpolation does not prevent an authorized quorum from misusing its authority or guarantee routing of funds to treasury accounts. ■

Mathematical Definition 9.2: Threshold Custodial Key Synthesis Operator

The joint custody of forfeited assets across multiple legal jurisdictions is mathematically defined through polynomial projection operators.

Definition 9.2.1 (Vandermonde Evaluation Operator). Let $\mathbf{x}_Q = [x_1, x_2, \dots, x_t]^T \in \mathbb{F}_q^t$ be the coordinates of the participating agencies. Define the Vandermonde evaluation matrix $\mathbf{V}_Q \in \mathbb{F}_q^{t \times t}$:

$$\left(\mathbf{V}_Q\right)_{i,j} = x_i^{j-1} \pmod{q}, \quad \forall i, j \in \{1, \dots, t\}$$

Definition 9.2.2 (Lagrange Weight Functional). The linear synthesis vector $\boldsymbol{\lambda}_Q \in \mathbb{F}_q^t$ that maps private shares $\mathbf{y}_Q = [y_1, y_2, \dots, y_t]^T$ to the underlying key is:

$$\boldsymbol{\lambda}_Q = \left(\mathbf{V}_Q^{-1}\right)^T \mathbf{e}_1$$

where $\mathbf{e}_1 = [1,0, \dots ,0]^T$ is the unit selection vector.

Definition 9.2.3 (Custodial Recovery Mapping). The authorization operator $\Psi_{custody}: \mathbb{F}_q^t \to \mathbb{F}_q$ evaluates as the inner product:

$$\Psi_{custody}\left(\mathbf{y}_Q\right) = \boldsymbol{\lambda}_Q^T \mathbf{y}_Q \equiv d_{seize} \pmod{q}$$

The synthesis vector is correct for the valid selected Vandermonde system. All ten three-of-five subsets in the fixture reconstruct, and an independent modular linear solve matches the Lagrange weights. Duplicate or zero coordinates are rejected. A modified share can instead reconstruct the wrong secret without inherent detection. The operator therefore establishes conditional recovery of a scalar, not a guarantee that international cooperation is always necessary for release or that all participants enforce a custody policy.

9.3 Theorem 9.3 Conditional Inclusion under Builder Priority Auctions

Theorem 9.3 (Conditional Builder Comparison). Let T_evader and T_seize be two modeled transactions whose state effects can be mutually exclusive. A builder may compare their feasible candidate payloads, including priority payments, direct payments, residual revenues and costs. A bid larger than the revenue of one competitor wins only in a restricted equal-residual, equal-cost two-choice model. It does not guarantee next-block inclusion, index zero or zero probability of the competing action. The source bid expression is retained below as an uncalibrated proposal:

$$B_{seize} \geq \Delta v_{evader} \cdot \mu_{bribe} + \epsilon$$

The expression's asset-value and multiplier terms do not establish the builder's opportunity costs or the existence of a feasible first-position payload. The inequality cannot supply the unconditional inclusion guarantee originally claimed. The required economic and execution assumptions are examined in the following retained auction discussion.

Proof

1. **Public Mempool Timing. A transaction propagates through a network with implementation-dependent delay. An exponential gossip-delay model would need calibration. Public transaction order is not universally determined by sorting one declared priority-fee field; builders face fee caps, dependencies, gas capacity and other sources of revenue. For a feasible EIP-1559 transaction with maxFee at least baseFee, the effective priority payment is:**

$$g_{\text{tip}} = \min(\text{maxPriorityFeePerGas}, \text{maxFeePerGas} - \text{baseFee})$$

A sender may attempt a higher-fee replacement where its transaction pool and protocol rules permit it, but replacement acceptance and eventual ordering are not guaranteed. A higher declared priority field can also be limited by the maximum-fee cap. The original front-running example is therefore a possible behavior, not a deterministic ordering rule.

2. **Builder Optimization. In the considered PBS-style model, a builder evaluates feasible payloads and its payment obligations to the proposer. Base fees are burned rather than received as producer fee revenue. A stylized objective therefore counts effective priority payments and direct revenues, less the relevant obligations and costs, subject to block and execution constraints:**

$$\text{maximize}_{\text{B}\in\text{feasible}} \left[\sum_{\text{T}\in\text{B}} \text{gas(T)} g_{\text{tip}}(\text{T}) + \text{direct(B)} - \text{bid}_{\text{validator}} - \text{costs(B)}\right]$$

3. **Private Bundle Conditions. A submitting party can request private routing and bundle constraints under the receiving service's actual supported semantics. These are conditions on acceptable execution, not proof that a qualifying payload exists or will be selected. The original first-position proposal is:**
   - Condition A: T_{seize} must be placed at execution index 0 of block H_{next}.
   - Condition B. Under the assumed service policy, a bundle failing its specified execution or position requirement is excluded from that candidate payload. This policy is an input to the model; a claim of no on-chain footprint for every related transaction would require the actual implementation and routing behavior to be established.

To incentivize the builder to select the enforcement bundle over the evader's public transaction, the authority includes a direct payment $\Delta\text{coinbase}_{direct} = B_{seize}$.

4. **Restricted Two-Choice Comparison. Let U_evader be the effective priority revenue from the competing transaction under the chosen fee model:**

$$U_{evader} = \text{gas}(T_{evader}) \cdot g_{tip,evader}$$

Suppose the two modeled actions are mutually exclusive in the relevant state. Let R_E and R_S denote all other revenue in their respective feasible payloads, and C_E and C_S the corresponding costs. Avoid including the competing transaction again in the residual revenue. The correct comparison is:

$$\Pi_{\text{E}} = \text{U}_{\text{evader}} + \text{R}_{\text{E}} - \text{C}_{\text{E}}$$
$$\Pi_{\text{S}} = \text{B}_{\text{seize}} + \text{R}_{\text{S}} - \text{C}_{\text{S}}$$

Only when R_E = R_S, C_E = C_S, both choices are feasible and no better alternative exists does preferring the second payload reduce to:

$$B_{seize} > U_{evader}$$

Setting a bid to the competing revenue plus a positive margin is thus an illustrative decision inside that restricted model. Any actual funding authority and budget are separate external premises:

$$B_{seize} = U_{evader} + \delta, \quad \delta > 0$$

A counterexample shows why the unrestricted conclusion fails. With gas capacity 10, let S use 10 gas and pay 11, E use 1 gas and pay 10, and a compatible X use 9 gas and pay 50. Although 11 > 10, E plus X pays 60 and beats S. A same-block prerequisite can also make first position impossible at any bid. The following ordering is therefore an additional condition for a chosen payload, not a consequence of the bid inequality:

$$\mathrm{idx}(T_{seize}) = 0 < \mathrm{idx}(T_{evader})$$

If both transactions are included, T_seize successfully executes first and changes the exact state required by T_evader, then the latter operation fails in that model. If T_seize reverts, consults the wrong state or leaves another path available, earlier ordering is insufficient. Private delivery and local profitability also do not establish that the receiving builder supplies the selected next block. These conditions replace the original deterministic capture and index-zero guarantee. ■

Mathematical Definition 9.3: Block Space Priority Contraction Functional

The preemptive seizure of flight-risk assets is formalized through order-ranking functionals over the block execution sequence.

Definition 9.3.1 (Transaction State Vector). Let $T \in \mathcal{T}$ be a valid transaction. Define its execution vector $\mathbf{v}_{tx}(T) = \left[\nu_{gas}, g_{tip}, \Delta_{bribe}, \sigma_{route}\right]^T$, where $\sigma_{route} \in \{0,1\}$ indicates private builder routing ($\sigma_{route} = 1$) versus public gossip propagation ($\sigma_{route} = 0$).

Definition 9.3.2 (Feasible Payload Ordering). Give transactions explicit IDs. A permutation orders the IDs of an already selected subset; the full optimization must also select the subset and impose execution, nonce, gas and precedence constraints. Use one indexing convention. The source ranking expression can be interpreted only after these choices are defined:

$$\pi^* = \arg\max_{\pi} \sum_{i=1}^{M} \mathcal{R}\left(T_{\pi(i)}, \pi(i)\right)$$

For a private bundle requesting first position, index 0 in zero-based notation corresponds to position 1 in one-based notation. Its effective priority and direct payment contribute revenue only within a feasible payload. A prerequisite “prepare” before “seize” prevents index zero regardless of payment. Neither a rank formula nor a private-route flag proves delivery, selection or finality.

Definition 9.3.3 (Ordering Indicator). Define the indicator as one only if both transactions are included and the first has lower rank, and zero otherwise. Rank itself is defined only on included transactions. This indicator measures order rather than successful recovery:

$$\Omega_{\mathrm{order}} = 1[\text{both included and } \mathrm{rank}(\mathrm{T_{seize}}) < \mathrm{rank}(\mathrm{T_{evader}})]$$

Effective preemption additionally requires successful execution and a state change that blocks the specified later operation. The indicator can equal one when the first transaction fails or changes irrelevant state. B_seize > U_evader alone therefore neither forces this indicator to one nor establishes asset capture. As a fee example, 100 gas with base fee 10, max fee 12 and max priority fee 5 gives effective priority revenue 200 and burned base fees 1,000; counting base fees as builder revenue would misstate the comparison.

Conclusion

Multi-chain architectures make retrospective accounting and attribution more demanding, but an event-based representation can organize the available evidence. The full spatiotemporal tensor, peeling-chain discussion, bridge model and custodial mapping remain useful analytical components when their information loss and assumptions are explicit. They do not deterministically collapse thousands of addresses into one legal person. Identity attribution requires supported ownership links, verified records and a statistical interpretation appropriate to the evidence, rather than a threshold score presented as certainty.

The liability framework likewise supplies conditional calculations. Documented lot histories can preserve basis across transfers, and topology alone does not prove a zero-basis default or classify every movement as a disposition. Under the original illustrative inputs, the seven FMVs still total $73,800,000 and the modeled debt is $26,568,000 against $14,600,000 of terminal liquidity. The resulting $11,968,000 shortfall is reproducible, but global insolvency requires a complete balance sheet. Exact geometric sums and conditional crossing criteria replace the original universal exponential-depth and forfeiture claims.

The enforcement models distinguish risk signals, valid administrative restrictions, feasible transaction inclusion and realized recovery. A token guard can block specified transfers while active, but may be reversible. A private bid does not guarantee first position, a selected next block or successful blocking. Correctly parameterized Shamir sharing supports threshold reconstruction under its field and randomness assumptions; full distributed signing and custody policy are additional mechanisms. These technical results do not establish the legal authorization, access or timing needed for a real intervention.

Corporate routing analogies remain part of the paper's discussion, with formal correspondence limited to cases where an explicit graph map preserves the required structure and costs. Mathematical homomorphism, isomorphism and legal equivalence are different claims. The revised framework supports testable reconstruction and accounting under stated assumptions, while recognizing unresolved calibration, implementation and legal inputs. The immutable records and verified identities available in a particular case can contribute evidence; they are not a mathematically infallible ledger of intent or a universal guarantee of total financial forfeiture.

Bibliography

- **Bank Secrecy Act (BSA) & FATF Guidelines**: 31 U.S.C. § 5318(l) and Financial Action Task Force (FATF) Recommendation 16, pertaining to Virtual Asset Service Provider (VASP) identification mapping and custodial compliance.
- **Internal Revenue Code (IRC) & Treasury Regulations:**
    - IRC § 482: Allocation of Income and Deductions Among Taxpayers (applied to transfer pricing elasticity mapping).
    - IRC § 1001: Determination of Amount of and Recognition of Gain or Loss (applied to atomic dispositions via intermediate swap topologies).
    - IRC § 1012(c) and Treasury Regulation § 1.1012-1(j): original references associated with the manuscript's basis and lot-identification discussion. Their scope, dates and application to a particular wallet or account are not established by the mathematical validation and require separate verification.
    - IRC § 6663: original reference for civil fraud penalties. The example assumes a 75% rate on the entire modeled base; eligibility and legal application are external premises.
    - IRC § 6861: original reference for jeopardy assessment procedures. Neither a risk indicator nor a computed deficit establishes the procedural conditions or authority to act.
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Additional technical references used in the validation: SEC 2, version 2.0, section 2.4.1 for secp256k1 parameters; BIP32 for derivation rules and vectors; Solidity storage-layout documentation for mapping encodings; EIP-140 for REVERT gas behavior; EIP-1559 for effective priority payments and burned base fees; RFC 9591 for FROST threshold Schnorr signing; Ethereum MPT documentation for path-aware proofs; and Uniswap v2 core reserve and cumulative-price update logic. These references support the stated technical corrections rather than the original universal legal conclusions.

Revision record

The full original sequence of nine phases, numbered definitions, derivations, simulation records and references is retained. Corrections are integrated at the affected locations. The following record identifies the main change in each phase; it is not a certification that finite tests establish all assumptions.

Phase 1

Causal ranking, complete flow accounting, finite-digit limits, conditional statistical bounds, metric properties and attribution uncertainty.

Phase 2

Accepted-basis inputs, event classification, exact geometric sums, conditional insolvency thresholds and component-specific penalty and interest treatment.

Phase 3

Positive-threshold stopping times, dimensionless risk inputs, reversible token guards, two-endpoint masks and feasible quantity-aware recovery.

Phase 4

Preserved source table and totals, typed schema, branch bookkeeping, explicit simple-interest convention and liquidity versus global net worth.

Phase 5

ECDSA acceptance conditions, non-hardened BIP32 scope and path-aware MPT verification without universal identity or synchrony claims.

Phase 6

Original ledger reconciliation, exact 2.6667% implied reduction and separation of bounded debt from a growing debt-to-assets ratio.

Phase 7

Explicit graph maps, allocation units, finite vertex elimination, corrected retention objectives and limits of corporate-policy analogies.

Phase 8

Transferable lot records, FIFO ordering, unknown versus zero basis, TWAP sensitivity and effective versus actual AMM reserves.

Phase 9

Fixed-width storage encoding, correct prime order, shares-only secrecy, separate signing protocols and conditional builder inclusion.

Validation scope: 72 findings supported under stated assumptions, 54 findings contradicted, 103 findings insufficiently specified. The scripts test mathematical and protocol fixtures, not current legal requirements, actual identity, authority or recoverability.